

## Types of Research

### I. Application Perspective

#### Theoretical (Basic) Research

- **Goal:** To expand knowledge without immediate practical application.
- **Example:** Investigating quantum mechanics principles.
- **Method:** Conceptual analysis, mathematical modeling, literature review.

#### Applied Research

- **Goal:** To solve specific, practical problems.
- **Example:** Developing a new app to improve mental health.
- **Method:** Combines theory with real-world testing.

### II. Aims of the research

Three types of research—**exploratory**, **descriptive**, and **explanatory**

#### Exploratory Research

- **Purpose:** To investigate a problem that is not clearly defined.
- **Goal:** Discover ideas, ask questions, and identify patterns.
- **Approach:** Flexible and unstructured.
- **Example:** Interviewing people to understand why a new product isn't selling well.
- **Use Case:** Early stages of research when little is known about the topic.

#### Descriptive Research

- **Purpose:** To describe characteristics of a population or phenomenon.
- **Goal:** Answer questions like *who*, *what*, *when*, *where*, and *how*.
- **Approach:** Structured and often quantitative.
- **Example:** A survey measuring customer satisfaction across different age groups.
- **Use Case:** When you want to paint a detailed picture of a situation or group.

#### Explanatory Research (also called Causal Research)

- **Purpose:** To explain why something happens.
- **Goal:** Identify cause-and-effect relationships.
- **Approach:** Often experimental or longitudinal.
- **Example:** Studying whether social media usage causes anxiety in teenagers.
- **Use Case:** When you want to understand the underlying reasons or mechanisms.

| Feature       | Exploratory          | Descriptive           | Explanatory           |
|---------------|----------------------|-----------------------|-----------------------|
| Main Question | What's going on?     | What is it like?      | Why does it happen?   |
| Structure     | Unstructured         | Structured            | Highly structured     |
| Data Type     | Qualitative          | Quantitative or mixed | Quantitative or mixed |
| Outcome       | Hypotheses, insights | Profiles, patterns    | Causal relationships  |

Each type builds on the other: **exploratory** research helps define the problem, **descriptive** research outlines the details, and **explanatory** research digs into the reasons.

**Research approach to study a phenomena and determine Causal Relationships among different factors**

It is done through **Experimental Research and Empirical Research**

### Experimental Research

Experimental research is about **manipulating variables** to test a hypothesis.

- **Definition:** Involves controlled testing to establish **cause-and-effect** relationships.
- **Methodology:** Uses lab experiments, control groups, and randomization.
- **Environment:** Conducted in **controlled settings**, like laboratory.
- **Data Type:** Primarily **quantitative**.
- **Goal:** To **prove or disprove** a specific hypothesis.
- **Example:** Testing a new drug by giving it to one group and a placebo to another.

| Feature              | Empirical Research         | Experimental Research  |
|----------------------|----------------------------|------------------------|
| Approach             | Observational              | Manipulative           |
| Control of Variables | Low                        | High                   |
| Setting              | Natural or real-world      | Controlled (e.g., lab) |
| Objective            | Describe or infer          | Establish causality    |
| Data Types           | Qualitative & Quantitative | Mostly Quantitative    |

So, in short: **empirical research observes in live situation**,  
while **experimental research manipulates variables factors to see their effect**.

### Empirical Research

- Empirical research is all about **observing the world as it is**. Observation and experiment are done in live environment. It seeks to answer questions using real-world evidence—whether through surveys, field studies, or experiments.
- **Definition:** Gathers knowledge through direct or indirect observation or experiment.
- **Methodology:** Includes surveys, case studies, field observations, and archival analysis.
- **Environment:** Typically conducted in **natural settings**.
- **Data Type:** Can be **qualitative** (e.g., interviews) or **quantitative** (e.g., statistics).
- **Goal:** To **describe, understand, or infer** patterns and relationships.
- **Example:** Studying consumer behavior by analyzing purchase data from retail stores.

### **Analytical Research:**

At its core, analytical research is about **breaking down complex information**, identifying patterns, and using logic and evidence to reach conclusions. It's less about the method (observation vs. experimentation) and more about the **critical thinking and structured analysis** behind it.

The research can be Longitudinal or/ and cross-sectional.

#### **Longitudinal Research**

- **Goal:** To study changes over time.
- **Example:** Tracking academic performance of students over 10 years.
- **Method:** Repeated observations of the same subjects.

#### **Causal-Comparative Research (Cross-section research)**

- **Goal:** To determine cause-effect relationships by comparing groups.
- **Example:** Comparing test scores of students taught with different methods.
- **Method:** No manipulation—uses naturally occurring groups.

Research can be **Qualitative** Research and/or **Quantitative** Research

| Qualitative Research                                                                                                | Quantitative Research                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Data: text, images, audio                                                                                           | Data: numerical, labels                                                                                                                          |
| Data collection: unstructured, flexible<br>Example: interview or survey with spontaneous open-ended questions       | Data collection: structured, standardized, sometimes subjective objective<br>Example: interview or survey with pre-defined close-ended questions |
| Usage: used in unexplored research topics and topics that are hard to quantify                                      | Usage: used for precision and conclusive                                                                                                         |
| Qualitative research uses a more unstructured and flexible approach, which works well in unexplored research fields | Quantitative research uses a more structured and standardized approach, which works well for zooming into details with high objectivity          |
| In well-researched fields, quantitative research tends to be more common                                            |                                                                                                                                                  |

### Example

Imagine a study testing whether listening to upbeat music improves creativity at work. Researchers split participants into two groups—one hears music, the other doesn't—and then measure their creative output. That's both **experimental** and **empirical**, because it involves:

- A controlled setup (experiment)
- Real-world data collection (empirical)

In short: **All experimental research is empirical**, but **not all empirical research is experimental**. Observational studies, for instance, are empirical but not experimental.

### Experimental vs. Empirical vs. Analytical

| Type of Research    | Definition                                                                          | Is Historical Data Used?                                     | Causal Inference Possible?                       |
|---------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------|
| <b>Experimental</b> | Involves controlled experiments where variables are manipulated to observe outcomes | Not typically (unless historical experiments were conducted) | Strong causal inference                          |
| <b>Empirical</b>    | Based on observed and measured phenomena; uses data to test hypotheses              | Yes, including historical data                               | With proper methods (e.g., regression, matching) |
| <b>Analytical</b>   | Focuses on breaking down concepts, theories, or models through logical reasoning    | Rarely uses raw data                                         | Not designed for causal testing                  |

Research is an art as well as a science.

### **Research Methods Vs Methodology:**

Research methods include all those techniques that are adopted for conducting research.

Research methodology is the way in which research problems are solved systematically. It is a science of doing research.

It is not only important for the researcher to know the research methods (techniques), but also the scientific approach called methodology.

### **Steps in Research Process**

Research process contains a series of closely related activities to carry out research.

1. Selection of Research Topic/ Area
2. Extensive Literature Survey
3. Define Research Problem
4. Formulation of Research Question, Objectives and Hypothesis
5. Preparing the Research Design
6. Define Population and Sampling
7. Data collection
8. Data Analysis
9. Hypothesis Testing
10. Generalization and Interpretation
10. Preparation of Report

### **Selection of Research Topic or Problem**

The research topic or problem should be practical, relatively important, feasible, ethically and politically acceptable. It is selected based on preliminary studies

### **Literature Review or Extensive Literature Survey**

After the selection of research problem, the second step is that of literature mostly connected with the topics. The availability of the literature may bring ease in the research. For this purpose academic journals, conference and govt. reports and library must be studied.

### **Define Research Problem**

- **What it is:** A specific issue, gap, or challenge within the topic that needs investigation.
- **Purpose:** Justifies why the research is necessary.
- **Example:** *Retail HR departments struggle to adapt to digital tools, leading to inefficiencies.*

**Formulation of Research Questions, Objectives and Making Hypothesis:** Define what is to be discovered or tested. Hypothesis means assumptions, which may be true, partially true, or wrong. The hypothesis is to draw the positive and

negative cause and effect aspects of a problem. Hypothesis narrows down the area of a research and keep a researcher on the right path.

### **Preparing the Research Design**

Choose methodology (qualitative, quantitative, or mixed). Decide on tools, techniques, and procedures for data collection. Detail plan on how to gather data for testing of hypothesis

Research design may be made, depend on the nature and purpose of the study. During Research Design the information about sources, skill, time and finance is taken into consideration.

### **Define Population and Sampling**

The researcher must select population and design a sample from where he/she can get data. The sample may be of two types:

1. Probability Sampling
2. Non-probability Sampling

### **Data collection**

Data collection is the most important work, is researcher. The collection of information must be containing on facts which is from the following two types.

**Primary Data Collection:** Primary data may be from the following.

1. Experiment
2. Questionnaire
3. Observation
4. Interview

**Secondary data collection:** it has the following categories:

1. Review of literature
2. Official and non-official reports
3. Library approach

### **Data Analysis**

To draw inference.

Data analysis may be divided into two main categories.

- Data Processing -- Data editing, Data classification, Data tabulation, Data presentation, Data measurement
- Data Exposition -- Description, Explanation, Narration, Conclusion/Findings, Recommendations/Suggestions

### **Hypothesis Testing**

Based on data analysis hypothesis is tested whether to accept or reject the hypothesis.

### **Generalization and Interpretation**

The acceptable hypothesis is used to arrive at the process of generalization or to make a theory. Some types of research have no hypothesis for which researcher uses data analysis for interpretation.

### **Preparation of Report**

Research report is a formal document, which has a fixed format.

# Formulation of Research Question

A researcher must find a problem and formulate /define it for doing research.

The terms—**research topic**, **problem**, **question**, and **objective**—are closely related but serve distinct roles in shaping a research project.

## 1. Research Topic

- **What it is:** The general area or subject you want to explore.
- **Purpose:** Sets the broad context for your study.
- **Example:** *Digital transformation in retail HR.*

## 2. Research Problem

- **What it is:** A specific issue, gap, or challenge within the topic that needs investigation.
- **Purpose:** Justifies why the research is necessary.
- **Example:** *Retail HR departments struggle to adapt to digital tools, leading to inefficiencies.*

## 3. Research Question

- **What it is:** A precise question that your study aims to answer.
- **Purpose:** Guides the direction and scope of your research.
- **Example:** *How does digital transformation affect employee performance in retail HR?*

## 4. Research Objective

- **What it is:** A clear, actionable goal that outlines what you intend to achieve.
- **Purpose:** Breaks down the research question into measurable steps.
- **Example:**
  - *To assess employee perceptions of digital transformation.*
  - *To identify barriers to adoption of digital tools in retail HR.*

## How They Connect

| Element          | Role in Research Flow                            |
|------------------|--------------------------------------------------|
| <b>Topic</b>     | Identifies the area of research. Start Research  |
| <b>Problem</b>   | Identifies the issues / problem within the topic |
| <b>Question</b>  | Frames the inquiry to address the problem        |
| <b>Objective</b> | Defines the steps to answer the question         |

## Research Question and Objectives:

### Example 1: Environmental Science

#### Research Question:

*How does microplastic pollution affect marine biodiversity in coastal ecosystems?*

#### Research Objectives:

- To quantify the concentration of microplastics in selected coastal regions.
- To identify the species most affected by microplastic ingestion.
- To analyze changes in reproductive and feeding behavior due to microplastic exposure.

## **Example 2: Chemical Engineering**

### **Research Question:**

*What is the efficiency of bio-based catalysts in converting waste cooking oil into biodiesel?*

### **Research Objectives:**

- To synthesize bio-based catalysts using agricultural waste.
- To compare the conversion rates of biodiesel using bio-based vs. conventional catalysts.
- To evaluate the environmental impact of the biodiesel production process.

## **Example 3: Mechanical Engineering**

### **Research Question:**

*How can additive manufacturing improve the thermal performance of heat exchangers?*

### **Research Objectives:**

- To design and fabricate heat exchanger prototypes using 3D printing.
- To measure thermal conductivity and heat transfer rates of printed components.
- To assess the mechanical durability of printed heat exchangers under operational stress.

## **Example 4: Biomedical Engineering**

### **Research Question:**

*Can wearable biosensors reliably detect early signs of dehydration in athletes?*

### **Research Objectives:**

- To develop a prototype wearable biosensor for hydration monitoring.
- To test sensor accuracy under various physical activity levels.
- To correlate biosensor data with clinical hydration markers.

Each objective breaks the question into measurable tasks, which helps to you build a structured and focused study.

## **WHAT IS A RESEARCH PROBLEM?**

Some difficulty is being faced and solution is not well known.

A research problem does exist if the following conditions are met:

(i) Problem must be real. There must be an individual (or a group or an organisation), to whom the problem can be attributed. The individual or the organisation, occupies an environment, which is defined by values of the uncontrolled variables.

(ii) There must be at least two courses of action, to be pursued. A course of action is defined by one or more values of the controlled variables. For example,

Problem → heart disease → Blood pressure, Heart failure, stroke

Course of action → Decide what food to eat or not eat

→ Decide what exercise to do or not do

→ Decide what medicines to take

(iii) There must be at least two possible outcomes, of which one should be preferable to the other. In other words, this means that there must be at least one outcome that the researcher wants, i.e., an objective.

Blood pressure outcome: low, normal, high, very high

(iv) The courses of action available must provide some chance of obtaining the objective. But outcome of a course of action is not well known.

(v) There must be some environment(s) to which the difficulty pertains.

- there are several factors which may make the problem complicated.
- The environment may change affecting the efficiencies of the courses of action or the values of the outcomes;
- the number of alternative courses of action may be very large;
- persons not involved in making the decision may be affected by it and react to it favourably or unfavourably.

All such elements (or at least the important ones) may be thought of in context of a research problem.

## **SELECTING THE PROBLEM**

Following points may be observed in selecting a research problem.

- Significant: Being faced in real life. Too narrow or too vague problems should be avoided.
- Novelty: Solution is not well known. Subject which is overdone should not be normally chosen.
- Feasible: To find solution with constraints. E.g. Controversial subject, unfamiliar area in which researcher has very little knowledge or expertise, time availability, costs, infrastructure, etc.

(The selection of a problem must be preceded by a preliminary study to ascertain its significance, novelty, and feasibility of finding a solution within available constraints.

## **DEFINING THE PROBLEM**

The problem to be investigated must be defined unambiguously for that will help to discriminate relevant data from the irrelevant ones. A proper definition of research problem will enable the researcher to be on the track.

Questions like: What data are to be collected? What characteristics of data are relevant and need to be studied? What relations are to be explored. What techniques are to be used for the purpose?

Answers to all such questions can be determined only when the research problem has been well defined.



# Steps to determine research question and objective

## 1. Explore and Identify a Broad Research Topic

- **Purpose:** Find an area that interests you and has academic or practical relevance.
- **How:**
  - Read recent articles, news, or journals.
  - Reflect on personal or professional experiences.
  - Consider societal or technological trends.
- **Example:** *Renewable energy technologies.*

## 2. Conduct a Literature Review

- **Purpose:** Understand what's already known and where gaps exist.
- **How:**
  - Search academic databases (Google Scholar, Scopus, etc.).
  - Identify recurring themes, debates, or unanswered questions.
- **Outcome:** Helps refine your topic and spot potential problems.

## 3. Define the Research Problem

- **Purpose:** Pinpoint a specific issue or gap within your topic.
- **How:**
  - Ask: *What's not working? What's missing? What needs improvement?*
  - Look for contradictions, inefficiencies, or unexplored areas.
- **Example:** *Solar panels lose efficiency in high humidity regions.*

## 4. Formulate the Research Question

- **Purpose:** Frame the problem as a question that guides investigation.
- **How:**
  - Use open-ended formats: *How, Why, What is the impact of...*
  - Ensure it's specific, measurable, and researchable.
- **Example:** *How does humidity affect the performance of solar panels in coastal regions?*

## 5. Set Research Objectives

- **Purpose:** Define what you aim to achieve—step-by-step goals.
- **How:**
  - Break the question into smaller, actionable tasks.
  - Use verbs like *analyze, evaluate, compare, design, measure.*
- **Example:**
  - *To measure solar panel efficiency under varying humidity levels.*
  - *To identify materials that perform better in humid conditions.*

## Summary Table

| Element           | Description                    | Example                                      |
|-------------------|--------------------------------|----------------------------------------------|
| <b>Topic</b>      | Broad area of interest         | Renewable energy technologies                |
| <b>Problem</b>    | Specific issue or gap          | Efficiency loss in humid climates            |
| <b>Question</b>   | Inquiry to address the problem | How does humidity affect solar panel output? |
| <b>Objectives</b> | Goals to answer the question   | Measure, analyze, and propose solutions      |

## Framing a Good Research Question

### 1. Clarity and Focus

- **Too broad:** *"How does technology affect education?"*  
→ Narrow it: *"How does AI-based tutoring impact math performance in middle school students?"*
- **Too narrow:** *"How does one specific app affect one student's test score?"*  
→ Broaden slightly to allow meaningful analysis.

### 2. Do a Deeper Literature Review

- Look for:
  - Gaps in existing research
  - Conflicting findings
  - Emerging trends
- Ask: *Has this question already been answered? If yes, can there be different approach having some advantage?*

### 3. Test It for Feasibility

Factors to be considered:

- Can the data needed be collect?
- Is the scope manageable within time and resources constraint?
- Is it measurable or answerable?

### 4. Get Feedback

- Talk to mentors, peers, or experts and people associated/ experiencing the problem.
- Ask: *Does this question make sense? Is it too vague or too complex?*

### 5. Refine the Language

- Use **precise terms**: Avoid jargon unless necessary.
- Make it **open-ended** (for qualitative) or **specific and measurable** (for quantitative).
- Example transformation:
  - Vague: *"Is solar energy good?"*
  - Refined: *"What are the economic and environmental impacts of solar energy adoption in rural India?"*

## Common Pitfalls in Research Question Formulation

### 1. Being Too Broad or Too Vague

A question like *“How does technology affect society?”* is too sweeping to investigate meaningfully.

This can be narrowed down to a specific technology, population, and impact—e.g., *“How does smartphone use affect sleep quality among college students?”*

### 2. Being Too Narrow or Trivial

A question like *“What is the average number of pencils used by third-grade students at a particular school during the month of August?”* lacks broader relevance.

### 3. Lack of Feasibility

Asking something that requires data that can’t be accessed or methods can’t be used.

### 4. Ambiguous Language

- **Problem:** Vague terms like “better,” “impact,” or “effective” without context.
- **Fix:** Use precise, measurable language.

### 5. Ignoring the Literature

- **Problem:** Asking a question that’s already been answered or lacks grounding in existing research.
- **Fix:** Do a thorough literature review to identify gaps, contradictions, or emerging trends.

### 6. Multiple Questions in One

- **Problem:** Trying to answer too many things at once—e.g., *“How does diet and exercise and sleep affect mental health?”*
- **Fix:** Focus on one variable or break it into separate studies.

### 7. Lack of Alignment with Objectives

- **Problem:** Your question doesn’t match your research goals or methods.
- **Fix:** Make sure your objectives clearly support answering the question.

A good research question is:

- **Feasible**
- **Interesting**
- **Novel**
- **Ethical**
- **Relevant**

## RESEARCH DESIGN

### Step 1: Consider Aims and Data Requirement

What is the research question?

What data need to be collected to answer the research question?

### Step 2: Consider aims and approach

What approach to adopt to collect the data?

Qualitative or Quantitative data

### Practical and ethical considerations when designing research

, one needs to consider [research ethics](#).

- How much time is needed to collect data?
- Feasibility to gain access needed to get the data (e.g., by travelling to a specific location or contacting specific people)?
- Necessary research skills (e.g., [statistical analysis](#) or interview techniques)?
- Ethical approval, If research involves people or animals

At each stage of the research design process, make sure that your choices are practically feasible.

### Types of quantitative research designs

Quantitative designs can be split into four main types.

Experimental

Experimental

- Used to test causal relationships
- Involves manipulating an independent variable and measuring its effect on a dependent variable
- Subjects are randomly assigned to groups
- Usually conducted in a controlled environment (e.g., a lab)

Observation

- Used to test whether (and how strongly) variables are related
- Variables are measured without influencing them

Descriptive

- Used to describe characteristics, averages, trends, etc
- Variables are measured without influencing them

### Types of qualitative research designs

Qualitative designs are flexible and less strictly defined.

Case study

- Detailed study of a specific subject (e.g., a place, event, organization, etc).
- Data can be collected using a variety of sources and methods.
- Focuses on gaining a holistic understanding of the case.

Ethnography

- Detailed study of the culture of a specific community or group.
- Data is collected by extended immersion and close observation.
- Focuses on describing and interpreting beliefs, conventions, social dynamics, etc.

Grounded theory

- Aims to develop a theory inductively by systematically analyzing qualitative data.

### **Step 3: Identify Population and Sampling Method**

A population is the entire group that for which researcher wants to draw conclusions about, while a sample is the smaller group representing the population.

Defining the population

For example,

- people from a specific demographic, region or background
- people with a certain job or medical condition
- users of a particular product
- Engg. Frequency range, voltage range

Sampling methods

Even with a narrowly defined population, it is difficult to get data on all entities.

There are two main approaches: probability sampling and non-probability sampling.

Probability sampling

- Sample is selected using random methods
- Mainly used in quantitative research
- Allows you to make strong statistical inferences about the population

Non-probability sampling

- Sample selected in a non-random way
- Used in both qualitative and quantitative research
- Easier to achieve, but chance of research bias

Case selection in qualitative research

In some types of qualitative designs, sampling may not be relevant.

For example, in a case study, the aim is to deeply understand a specific context, not to generalize to a population.

Careful selection of cases relevant to research question.

### **Step 4: Choose data collection methods**

Data collection methods are ways of directly measuring variables and gathering information. They allow first-hand knowledge and insights into research problem.

**Survey methods**

Surveys allow you to collect data about opinions, behaviors, experiences, and characteristics by asking people directly. There are two main survey methods:

Questionnaires      Interviews

**Observation methods**

Observational studies allow data collect real life situation without causing obstruction.

Observations may be conducted in real time, taking notes as you observe, or you might make audiovisual recordings for later analysis. They can be qualitative or quantitative.

**Other methods of data collection**

There are many other ways you might collect data depending on field and topic.

- Media & communication: Collecting a sample of texts (e.g., speeches, articles, or social media posts) for data on cultural norms and narratives
- Psychology: Using technologies like neuroimaging, eye-tracking, or computer-based tasks to collect data on things like attention, emotional response, or reaction time

- Education: Using tests or assignments to collect data on knowledge and skills
- Physical sciences: Using scientific instruments like weight, blood pressure, or chemical composition
- Video and image for data processing

### **Secondary data**

Data that is already available at various sources.—for example, datasets from government surveys or previous studies on your topic.

## **Step 5: Plan Data Collection Procedures**

Plan exactly how data collection methods be used to collect data in a consistent, accurate, and unbiased manner.

### **Operationalization**

#### **Survey**

What questions to ask and what range of responses to be offered? -> Design of Questionnaire

Example

To measure teachers' satisfaction with online learning tools, you could create a questionnaire with a 5-point rating scale.

#### **Observation**

Some variables, like height or age, are easily measured. But often with more abstract concepts, like satisfaction, anxiety, or competence. Operationalization means turning these fuzzy ideas into measurable indicators.

Example

To measure student participation in an online course, you could record the number of times students ask and answer questions.

#### **Experiment**

-> Design of experimental setup

### **Reliability and validity**

Reliability means your results can be consistently reproduced, while validity means that you're actually measuring the concept you're interested in.

### **Sampling procedures**

How sampling to be done so the it is easy to carry out and is representative of the population

### **Data management**

It's also important to create a data management plan for organizing and storing your data.

## **Step 6: Decide on Data Analysis Strategies**

Raw Data Cannot answer the Research Question. Data Has To be analysed to draw inference.

Quantitative data analysis

In quantitative research, use some form of statistical analysis.

**Descriptive statistics:** Summarize sample data.

The distribution of the data (e.g., the frequency of each score on a test)

The central tendency of the data (e.g., the mean to describe the average score)

The variability of the data (e.g., the standard deviation to describe how spread out of scores)

The specific calculations depend on the level of measurement of your variables.

**Inferential Statistics:** Make estimates about the population based on sample data.

Test hypotheses about a relationship between variables.

Regression and correlation tests look for associations between two or more variables, while comparison tests (such as t tests and ANOVAs) look for differences in the outcomes of different groups.

**Qualitative data analysis**

Identify patterns, extract the parts that are most relevant and interpret its meanings to the research question.

# Design of Questionnaire

-- is both an art and a science

—it's about asking the right questions in the right way to get meaningful, reliable data.

## Step-by-Step Guide to Questionnaire Design

### 1. Define the Objective

- What is to be learnt or measured?
- Frame clear research questions or hypotheses.

### 2. Identify Target Respondents

- Who will be answering? (e.g., customers, students, professionals)
- Consider their language: literacy level, and familiarity with the topic.

### 3. Choose the Question Types

| Type           | Description                           | Use Case Example          |
|----------------|---------------------------------------|---------------------------|
| Closed-ended   | Multiple choice, Likert scale, Yes/No | Quantitative analysis     |
| Open-ended     | Free text responses                   | Qualitative insights      |
| Rating/Ranking | Rate or rank items                    | Preferences, satisfaction |
| Demographic    | Age, gender, location, etc.           | Segmenting respondents    |

### 4. Craft Clear and Neutral Questions

- Keep questions concise and jargon-free.
- Avoid leading or biased wording.

Example: Instead of “How amazing was our service?”

→ “How satisfied were you with our service?”

### 5. Sequence and Flow

- Start with easy, engaging questions.
- Group similar topics together.
- End with optional open-ended questions (e.g., remark, opinions).

### 6. Pilot Test the Questionnaire

- Run a small test with a sample group.
- Check for clarity, timing, and technical issues.

### 7. Finalize and Format

- Use consistent formatting (fonts, spacing).
- Include front matter, and instructions
- Ensure anonymity/confidentiality if needed.



**Validity** in the context of a questionnaire refers to how well the instrument measures what it is intended to measure. In other words, does the questionnaire actually capture the concept or construct being studied?

### **Types of Validity in Questionnaires**

| <b>Type</b>                                   | <b>What It Assesses</b>                                                                                                                                   | <b>Example</b> |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Content Validity</b>                       | Whether the questionnaire fully covers <b>all aspects</b> of the concept. e.g. A survey on job satisfaction includes items on pay, work environment, etc. |                |
| <b>Face Validity</b>                          | Whether the questions <b>appear relevant</b> and appropriate at face value.                                                                               |                |
| <b>Construct Validity</b>                     | Whether the questionnaire <b>truly measures the theoretical construct</b> . Scale measuring "anxiety" correlates with "stress".                           |                |
| <b>Convergent &amp; Discriminant Validity</b> | Subtypes of construct validity—do related constructs correlate (convergent) and                                                                           |                |

| Type               | What It Assesses                                                           | Example                                                                          |
|--------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|                    | unrelated ones not (discriminant)?                                         | A scale measuring creativity correlates with innovation but not with aggression. |
| Criterion Validity | Whether the questionnaire correlates with an external benchmark or outcome |                                                                                  |

### How to Ensure Validity in Practice

- **Define constructs clearly** before writing questions.
- **Consult experts** to review items for content and face validity.
- **Pilot test** questionnaire and collect feedback.
- Use **Exploratory Factor Analysis (EFA)** or **Confirmatory Factor Analysis (CFA)** to test construct validity.
- Compare results with **external measures** for criterion validity.

**Testing the validity of a questionnaire** is all about ensuring that instrument truly measures what it's supposed to measure.

#### 1. Content Validity

- Create a **blueprint** mapping each question to specific dimensions of the construct.
- Get **expert reviews** from subject matter specialists.
- Use a **Content Validity Index (CVI)**: Experts rate each item for relevance; calculate proportion of agreement.

#### 2. Face Validity

Does the questionnaire appear valid to respondents and

- Share the questionnaire with a small group of target users.

- Ask for feedback on clarity, relevance, and appropriateness.
- This is subjective but useful for refining wording and tone.

### 3. Construct Validity

Does the questionnaire measure the theoretical construct?

- Check **convergent validity** (correlates with similar constructs) and **discriminant validity** (does not correlate with unrelated constructs).

### 4. Criterion Validity

Does the questionnaire correlate with an external benchmark?

- **Concurrent validity:** Compare with an established instrument measuring the same construct.
- **Predictive validity:** See if scores predict future outcomes (e.g., performance, behaviour).

**Reliability in a questionnaire** refers to its **consistency**—whether it produces stable and repeatable results under consistent conditions. If you administer the same questionnaire to similar respondents at different times (or in different formats), a reliable instrument will yield similar outcomes.

#### Types of Reliability in Questionnaires

| Type                           | What It Measures                                         | Example                                                                |
|--------------------------------|----------------------------------------------------------|------------------------------------------------------------------------|
| <b>Internal Consistency</b>    | How well items within a scale measure the same construct | All items in a “stress” scale should correlate well with each other    |
| <b>Test-Retest Reliability</b> | Stability of responses over time                         | Administer the same questionnaire twice to the same group, weeks apart |
| <b>Inter-Rater Reliability</b> | Agreement between different raters or observers          | Two reviewers score open-ended responses similarly                     |
| <b>Split-Half Reliability</b>  | Consistency between two halves of the questionnaire      | Compare scores from odd vs. even items                                 |

#### How to Test Reliability

- **Cronbach’s Alpha** (for internal consistency):
  - Values range from 0 to 1

- $\geq 0.7$  is generally acceptable
- **Pearson Correlation** (for test-retest or split-half):
  - Measures linear relationship between two sets of scores
- **Cohen's Kappa** (for inter-rater reliability):
  - Adjusts for chance agreement between raters

Cronbach's Alpha ( $\alpha$ ) is a statistic used to measure the **internal consistency** or **reliability** of a set of items in a questionnaire. In simpler terms, it tells you how well a group of questions work together to measure the same underlying concept.

### What It Really Means

- If questionnaire has multiple items intended to assess a single construct (like motivation, anxiety, or satisfaction), Cronbach's Alpha checks whether those items are **correlated** and **consistent**.
- It's expressed as a value between **0 and 1**:
 

|                                                |                                                  |
|------------------------------------------------|--------------------------------------------------|
| $\alpha \geq 0.9 \rightarrow$ Excellent        | $0.8 \leq \alpha < 0.9 \rightarrow$ Good         |
| $0.7 \leq \alpha < 0.8 \rightarrow$ Acceptable | $0.6 \leq \alpha < 0.7 \rightarrow$ Questionable |
| ◦ $\alpha < 0.6 \rightarrow$ Poor              |                                                  |

### Formula (for the curious)

$$\alpha = \frac{N \cdot \bar{c}}{\bar{v} + (N - 1) \cdot \bar{c}}$$

Where:

- $(N)$  = number of items
- $(\bar{c})$  = average covariance between item pairs
- $(\bar{v})$  = average variance of each item

### Cronbach's Alpha calculation in Excel

Let's say we have a 4-item questionnaire measuring "User Satisfaction" and responses from 10 participants. Set up your Excel sheet like this:

#### Respondent Q1 Q2 Q3 Q4

|   |   |   |   |   |
|---|---|---|---|---|
| 1 | 3 | 4 | 3 | 5 |
| 2 | 2 | 3 | 2 | 4 |
| 3 | 4 | 5 | 4 | 5 |
| 4 | 3 | 3 | 3 | 4 |

### Respondent Q1 Q2 Q3 Q4

|    |   |   |   |   |
|----|---|---|---|---|
| 5  | 2 | 2 | 2 | 3 |
| 6  | 4 | 4 | 4 | 5 |
| 7  | 3 | 4 | 3 | 4 |
| 8  | 2 | 3 | 2 | 3 |
| 9  | 4 | 5 | 4 | 5 |
| 10 | 3 | 4 | 3 | 4 |

### Step 2: Calculate Item Variances

Use =VAR.S(range) for each question column:

- Q1: =VAR.S(B2:B11) = 0.6667
- Q2: =VAR.S(C2:C11) = 0.9
- Q3: =VAR.S(D2:D11) = 0.6667
- Q4: =VAR.S(E2:E11) = 0.6222

Sum of item variances = 2.8555

### Step 3: Calculate Total Score per Respondent

Add a new column "Total":

=SUM(B2:E2)

Drag down for all rows.

### Step 4: Calculate Variance of Total Scores

Use: =VAR.S(F2:F11) Let's say the result is: 10.3222

### Step 5: Apply Cronbach's Alpha Formula

$$\alpha = \frac{N}{N-1} \left( 1 - \frac{\sum \text{Item Variances}}{\text{Total Score Variance}} \right)$$

In Excel:

= (4 / (4 - 1)) \* (1 - (2.8555 / 10.3222))

Result:  $\alpha \approx 0.9645$

### Interpretation

- If  $\alpha \geq 0.7 \rightarrow$  Acceptable reliability
- If  $\alpha \geq 0.8 \rightarrow$  Good
- If  $\alpha \geq 0.9 \rightarrow$  Excellent

**Inter-rater reliability** becomes especially relevant when a questionnaire involves **subjective judgments**—where multiple evaluators interpret responses based on their own understanding.

### **Example: Evaluating Open-Ended Responses in a Job Interview Questionnaire**

Imagine you're designing a questionnaire for hiring customer service representatives. One section includes open-ended questions like:

“Describe a time you handled a difficult customer.”

Now, three HR professionals independently rate each response on a scale of 1 to 5 for:

- Empathy
- Problem-solving
- Communication clarity

Here's where **inter-rater reliability** matters:

- If Rater A gives a response a 5, Rater B gives it a 2, and Rater C gives it a 4, the inconsistency suggests poor inter-rater reliability.
- But if all three raters consistently score similar responses similarly, it shows the rating system is **stable and objective**.

### **Why It's Important**

- Ensures fairness in scoring
- Validates the consistency of your measurement tool
- Helps refine rating criteria or training for raters

You could use **Cohen's Kappa** or **Intraclass Correlation Coefficient (ICC)** to quantify this reliability, depending on the scale and number of raters.

## MEASUREMENT SCALES

Measurements figures permit statistical analysis and communication of measurement results.

Scaling is an extension of measurement. Scaling involves creating a continuum on which measurements are made.

### Types of Measurement Scales

There are four types of measurement scales

1. Nominal Scale,
2. Ordinal Scale,
3. Interval Scale      &      4. Ratio Scale.

The choice of the measurement scale has implications for the statistical technique to be used for data analysis.

**Nominal Scale:** This is the lowest level of measurement. Here, numbers are assigned for the purpose of identification of the objects. Any object which is assigned a higher number is in no way superior to the one which is assigned a lower number. The objects are divided into mutually exclusive and collectively exhaustive categories. Each number is assigned to only one object and each object has only one number assigned to it.

Example:

 What is your religion?

1. Hinduism
2. Sikhism
3. Christianity
4. Islam
5. Any other, (please specify)

Any religion which is assigned a higher number is in no way superior or inferior to the one which is assigned a lower number. The assignment of numbers is only for the purpose of identification.

Nominal scale measurements are used for identifying attributes, such as food habits (vegetarian or non-vegetarian), gender (male/female), caste, respondents, marital status, brands, etc.

The assigned numbers cannot be added, subtracted, multiplied or divided.

The only arithmetic operations that can be carried out are the count of each category.

Therefore, a frequency distribution table can be prepared for the nominal scale variables and mode of the distribution can be worked out.

**Ordinal scale:** This is the next higher level of measurement than the nominal scale measurement.

An ordinal scale measurement tells whether an object has more or less of characteristics than some other objects. However, it cannot answer how much more or how much less.

Example:

 Rank the following attributes while choosing a restaurant for dinner. The most important rank one, the next important rank of 2 and so on.

Food quality [ ]      Prices [ ]      Menu variety [ ]  
Ambience [ ]      Service [ ]

In the ordinal scale, the assigned ranks cannot be added, multiplied, subtracted or divided. One can compute median, percentiles and quartiles of the distribution. The other major statistical analysis which can be carried out is the rank order correlation coefficient. All the statistical techniques which are applicable in the case of nominal scale measurement can also be used for the ordinal scale measurement.

**Interval scale:** The interval scale measurement is the next higher level of measurement. It takes care of the limitation of the ordinal scale measurement where the difference between the score on the ordinal scale does not have any meaningful interpretation. In the interval scale the difference of the score on the scale has meaningful interpretation. It is assumed that the respondent is able to answer the questions on a continuum scale. The mathematical form of the data on the interval scale may be written as

$$Y = a + b X \quad \text{where } a \neq 0$$

In the interval scale, the difference in score has a meaningful interpretation while the ratio of the score on this scale does not have a meaningful interpretation.

This can be seen from the following interval scale question:

 How likely are you to buy a new designer carpet in the next six months?

| .       | Very unlikely | Unlikely | Neutral | Likely | Very likely |
|---------|---------------|----------|---------|--------|-------------|
| Scale A | 1             | 2        | 3       | 4      | 5           |
| Scale B | 0             | 1        | 2       | 3      | 4           |
| Scale C | -2            | -1       | 0       | 1      | 2           |

Suppose a respondent ticks the response category 'likely' and another respondent ticks the category 'unlikely'. If we use any of the scales A, B or C, we



note that the difference between the scores in each case is 2. Whereas, when the ratio of the scores is taken, it is 2, 3 and -1 for the scales A, B and C respectively.

Therefore, the ratio of the scores on the scale does not have a meaningful interpretation. The following are some examples of interval scale data.

 How important is price to you while buying a car?

Least important, Unimportant, Neutral, Important, Most important (1 2 3 4 5)

 How do you rate the work environment of your organization?

Very good, Good, Neither good nor bad, Bad, Very bad (5 4 3 2 1)

 How expensive is the restaurant 'XYZ'?

Extremely expensive, Definitely expensive, Somewhat expensive, Somewhat inexpensive, Definitely inexpensive, Extremely inexpensive (1 2 3 4 5 6)

The numbers on this scale can be added, subtracted, multiplied or divided. One can compute arithmetic mean, standard deviation, correlation coefficient, regression analysis and factor analysis. All the techniques applicable for the ordinal and the nominal scale data can also be used for interval scale data.

**Ratio scale:** This is the highest level of measurement and takes care of the limitations of the interval scale measurement.

The ratio scale measurement can be converted into interval, ordinal and nominal scale. But the other way round is not possible. The mathematical form of the ratio scale data is given by

$$Y = b X.$$

Examples of the ratio scale data are weight, distance travelled, income and sales of a company, to mention a few.

## CLASSIFICATION OF SCALES

One of the ways of classifications of scales is in terms of the number of items in the scale. Based upon this, the following classification may be proposed:

### I. Single Item vs Multiple Item Scale

**Single item scale:** In the single item scale, there is only one item to measure a given construct. Example:

Consider the following question:

 How satisfied are you with your current job?

Very Dissatisfied, Dissatisfied, Neutral, Satisfied, Very satisfied

The problem with the above question is that there are many aspects to a job, like pay, work environment, rules and regulations, security of job and relationship with the seniors. The respondent may be satisfied on some of the factors but may not on others. By asking a question as stated above, it will be difficult to analyse the problem areas. To overcome this problem, a multiple item scale is proposed.

**Multiple item scale:** In multiple item scale, there are many items (factors) that affect the construct that the researcher is trying to measure. Each of the item forms some part of the construct (satisfaction) which the researcher is trying to measure. Example

■ How satisfied are you with the pay you are getting on your current job?

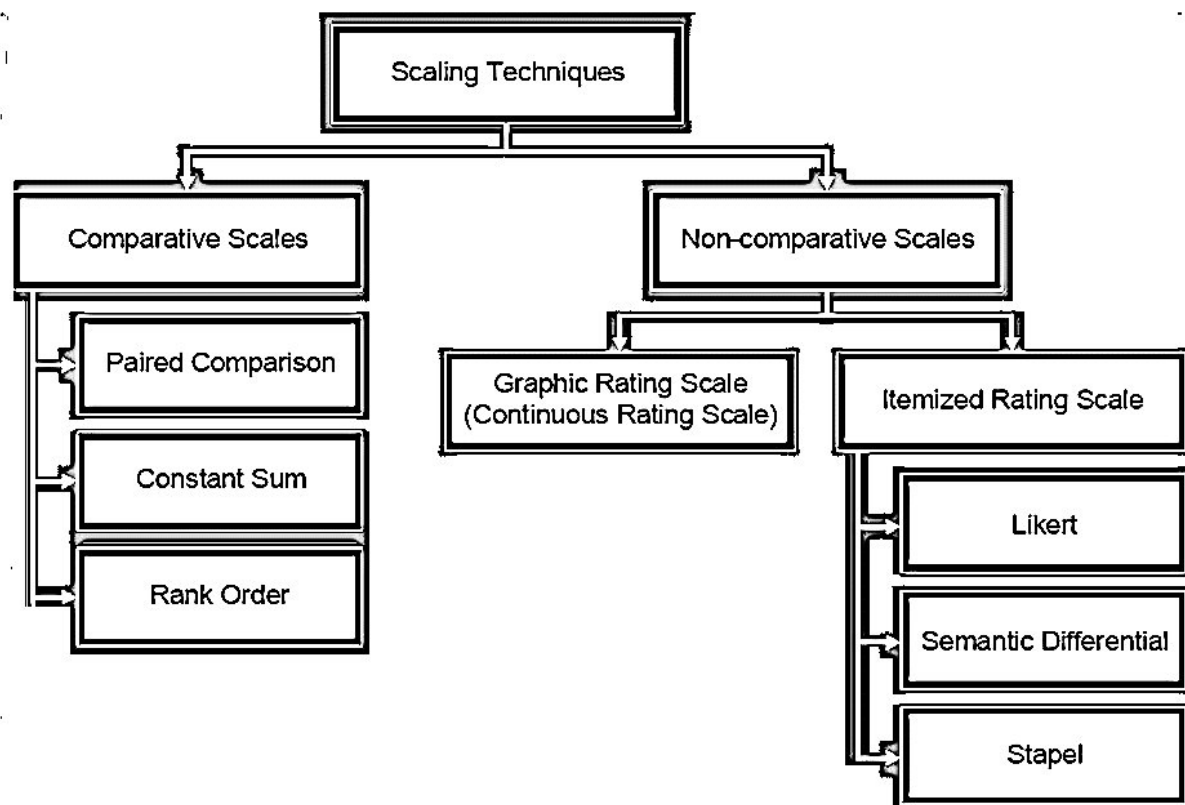
Very dissatisfied, Dissatisfied, Neutral, Satisfied, Very satisfied

■ How satisfied are you with the rules and regulations of your organization?

Very dissatisfied, Dissatisfied, Neutral, Satisfied, Very satisfied

## II. Comparative vs Non-comparative Scales

The scaling techniques used in research can also be classified into comparative and non-comparative scales.



### i) Comparative scales

In comparative scales it is assumed that respondents make use of a standard frame of reference before answering the question. For example:

**Paired comparison scales:** Here a respondent is presented with two objects and is asked to select one according to whatever criterion he or she wants to use.

Example:

 Please rate Domino's in comparison to Pizza Hut on the basis of your satisfaction level on the following parameters:

(1 = Much Worse, 2 = Worse, 3 = Same, 4 = Better, 5 = Much Better.)

- a. Variety of menu options [ ]      b. Value for money [ ]  
c. Speed of service (delivery time) [ ]      d. Food quality [ ]

The resulting data from this scale is ordinal in nature.

**Rank Order Scale:** In the rank order scaling, respondents are presented with several objects simultaneously and asked to order or rank them according to some criterion. Consider, for example the following question:

 Rank the following soft drinks in order of your preference, the most preferred soft drink should be ranked one, the second most preferred should be ranked two, and so on.

- Coke [ ]      Pepsi [ ]      Limca [ ]      Sprite [ ]  
Mirinda [ ]      Fanta [ ]      Seven Up [ ]

Like paired comparison, this approach is also comparative in nature. The problem with this scale is that if a respondent does not like any of the above mentioned soft drink and is forced to rank them in the order of his choice.

The rank order scaling results in the ordinal data.

**Constant Sum Rating Scale:** In constant sum rating scale, the respondents are asked to allocate a total of 100 points between various objects and brands. The respondent distributes the points to the various objects in the order of his preference. Example:

 Allocate a total of 100 points among the various schools into which you would like to admit your child. The points should be allocated in such a way that the sum total of the points allocated to various schools adds up to 100.

- |                   |                   |
|-------------------|-------------------|
| DPS               | Mother's Integral |
| Sai International | DAV Public School |

This type of data is not only comparative in nature but could also result in ratio scale measurement.

## ii) Non-comparative scales

In the non-comparative scales, the respondents do not make use of any frame of reference before answering the questions. The resulting data is generally assumed to be interval or ratio scale.

The non-comparative scales are divided into two categories, namely, the Graphic Rating scales and the Itemized rating scales.

**Graphic rating scale:** This is a continuous scale, also called graphic rating Scale. In the graphic rating scale the respondent is asked to tick his preference on a graph. Example:

 Please put a tick mark (📍) on the following line to indicate your preference for fast food.



To measure one has to measure the distance from the extreme left to the position where a tick mark has been put.

## Itemized rating scale

In the itemized rating scale, the respondents are provided with a scale that has a number of brief descriptions associated with each of the response categories. There are certain issues that should be kept in mind while designing the itemized rating scale. These issues are:

**Number of categories to be used:** There is no hard and fast rule as to how many categories should be used in an itemized rating scale. However, it is a practice to use five or six categories. Some researchers are of the opinion that more than five categories should be used in situations where small changes in attitudes are to be measured. There are others that argue that the respondents would find it difficult to distinguish between more than five categories.

**Odd or even number of categories:** It has been a matter of debate among the researchers as to whether odd or even number of categories are to be used. By using even number of categories the scale would not have a neutral category and the respondent will be forced to choose either the positive or the negative side of the attitude. If odd numbers of categories are used, the respondent has the freedom to be neutral if he wants to be so.

**Balanced versus unbalanced scales:** A balanced scale is the one which has equal number of favourable and unfavourable categories. The following is the example of a balanced scale:

🎬 How important is price to you in buying a new car?

Very important, Relatively important, Neither important nor unimportant, Relatively unimportant, Very unimportant

In this question, there are five response categories, two of which emphasize the importance of price and two others that do not show its importance. The middle category is neutral.

The following is the example of the unbalanced scale.

🎬 How important is price to you in buying a new car?

Extremely important, Important, Somewhat important, Unimportant  
In this question there are four response categories that are skewed towards the importance given to the price, whereas one category is for the unimportant side. Therefore, this question is an unbalanced question.

**Forced versus non-forced scales:** In the forced scale, the respondent is forced to take a stand, whereas in the non-forced scale, the respondent can be neutral if he/she so desires. The argument for a forced scale is that those who are reluctant to reveal their attitude are encouraged to do so with the forced scale. Paired comparison scale, rank order scale and constant sum rating scales are examples of forced scales.

**Likert scale:** This is a multiple item agree-disagree five-point scale. The respondents are given a certain number of items (statements) on which they are asked to express their degree of agreement/disagreement. This is also called a summated scale because the scores on individual items can be added together to produce a total score for the respondent. An assumption of the Likert scale is that each of the items (statements) measures some aspect of a single common factor, otherwise the scores on the items cannot legitimately be summed up. In a typical research study, there are generally 25–30 items on a Likert scale.

To construct a Likert scale to measure a particular construct, a large number of statements pertaining to the construct are listed. These statements could range from 80–120. The identification of the statements is done through exploratory research which is carried out by conducting a focus group, unstructured interviews with knowledgeable people, literature survey, analysis of case studies and so on.

Suppose we want to assess the image of a company. As a first step, an exploratory research may be conducted by having an informal interview with

the customers, and employees of the company. The general public may also be contacted.

Suppose the number of statements to measure the constructs is 100 in number. Now samples of representative respondents are asked to state their degree of agreement/disagreement on those statements.

Table gives a few statements to assess the image of the company.

Likert Scale Statements to Measure the Image of the Company

Strongly Disagree (SD)    Disagree (D)

Neither Agree nor Disagree (N)    Agree (A)    Strongly Agree (SA)

1. The company makes quality products \_\_\_\_
2. It is a leader in technology \_\_\_\_
3. It doesn't care about the general public. \_\_\_\_\_
4. The company leads in R&D to improve products. \_\_\_\_\_
5. The company is not a good paymaster. \_\_\_\_
6. The products of the company go through stringent quality tests. \_\_\_\_
7. The company has not done anything to curb pollution. \_\_\_\_
8. It does not care about the community near its plant. \_\_\_\_
9. The company's stocks are good to buy or own. \_\_\_\_
10. The company does not have good labour relations. \_\_\_\_\_

The scale contains statements' some of which are favourable to the construct we are trying to measure and some are unfavourable to it.

Example, statements numbering 1, 2, 4, 6 and 9 are favourable statements, whereas the remaining are unfavourable statements. Generally, in a Likert scale, there is an approximately equal number of favourable and unfavourable statements. After the scale is administered, numerical values are assigned to the responses. The rule is that a 'strongly agree' response for a favourable statement should get the same numerical value as the 'strongly disagree' response of the unfavourable statement.

What is the reason for having a mixture of favourable and unfavourable statements in a Likert scale?

- The responses by the respondent should not become monotonous.
- It serves Validity and Reliability

For 10 statements maximum score is 50 and minimum is 10.

Now if there are 100 statements, the score on the image of the company can be worked out for each respondent by adding his/her scores on the 100

statements. The minimum score for each respondent will be 100, whereas the maximum score would be 500.

A typical Likert scale comprises about 25–30 statements. In order to select 25 statements from the 100 statements, we need to discard some of them. The rule behind discarding the statements is that those items that are non-discriminating should be removed.

## MEASUREMENT ERROR

Measurement error occurs when the observed measurement on a construct or concept deviates from its true values. The following is a list of the sources of measurement errors.

■ There are factors like mood, fatigue and health of the respondent which may influence the observed response while the instrument is being administered. The other factors could be education, job, awareness of topic and reluctance to express an opinion.

■ At times, the errors may be committed at the time of coding, entering of data from questionnaire to the spreadsheet on the computer and at the tabulation stage. The other reasons could be defective instrument for data collection like lengthy and ambiguous questionnaire with leading questions (suggestive responses) in the instrument.

The observed measurement in any research need not be equal to the true measurement. The observed measurement can be written as

$$O = T + S + R \quad \text{Where,}$$

O = Observed measurement

T = True score

S = Systematic error

R = Random error

It may be noted that the total error consists of two components—systematic error and random error. Systematic error causes a constant bias in the measurement.

Suppose there is a weighing scale that weighs 50 gm less for every one kg of product being weighed. The error would consistently remain the same irrespective of the kind of product and the time at which product is weighed. Random error on the other hand involves influences that bias the measurements but are not systematic.

Suppose we use different weighing scales to weigh one kg of a product and if systematic error is assumed to be absent, we may find that recorded weights

may fall within a range around the true value of the weight, thereby causing random error.

### **Criteria for Good Measurement**

There are three criteria for evaluating measurements:

(1) Reliability, (2) Validity and (3) Sensitivity.

#### **1. Reliability**

**Test-retest reliability:** In this method, repeated measurements of the same person or group using the same scale under similar conditions are taken. A very high correlation between the two scores indicates that the scale is reliable.

**Split-half reliability method:** This method is used in the case of multiple item scales. Here the number of items is randomly divided into two parts and a correlation coefficient between the two is obtained. A high correlation indicates that the internal consistency of the construct leads to greater reliability.

#### **2. Validity**

The validity of a scale refers to the question whether we are measuring what we want to measure. Validity of the scale refers to the extent to which the measurement process is free from both systematic and random errors.

**Content validity:** This is also called face validity. It involves subjective judgement by an expert for assessing the appropriateness of the construct. For example, to measure the perception of a customer towards an Airlines, a multiple item scale is developed. However researcher might have missed some important item.

In order to judge the content validity of these items, a set of experts may be requested to examine the representativeness of the 15 items.

**Predictive validity:** This involves the ability of a measured phenomena at one point of time to predict another phenomenon at a future point of time. If the correlation coefficient between the two is high, the initial measure is said to have a high predictive ability. E.g. CAT and MAT score, UPSC and OPSC

#### **3. Sensitivity**

Sensitivity refers to an instrument's ability to accurately measure the variability in a concept. A dichotomous response category such as agree or disagree does not allow the recording of any attitude changes. A more sensitive measure with numerous categories on the scale may be required.



than a single-item scale.

## NOTES

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Attitude Measurement  
and Scaling

Check Your Progress

7. What are systematic errors?

8. Name the method in which the number of items is randomly divided into two parts and a correlation coefficient between the two is obtained.

## 5.5 ANSWERS TO CHECK YOUR PROGRESS

### QUESTIONS

1. One of the limitation of the nominal scale measurements is that we cannot say whether the assigned number to an object is higher or lower than the one assigned to another option. The ordinal scale measurement takes care of this limitation.

2. The highest level of measurement is the ratio scale.
3. The intention component of attitude also called the behavioural component, reflects a predisposition to an action by reflecting the consumer's buying or purchase intention.
4. The Q-sort technique makes use of the rank order procedure in which objects are sorted into different piles based on their similarity with respect to certain criterion.
5. Some examples of forced scales include paired comparison scale, rank order scale and constant sum rating scales.
6. The Likert scale is also called as summated scale because the scores on individual items can be added together to produce a total score for the respondent.
7. Systematic errors are one of the components of total error which causes a constant bias in the measurement.
8. Split-half reliability method is the method in which the number of items is randomly divided into two parts and a correlation between the two is obtained.

## 5.6 SUMMARY

 Measurement means the assignment of numbers or other symbols to the characteristics of certain objects. Scaling is an extension of measurement. Scaling involves creating a continuum on which measurements on the objects are located. There are four types of measurement scales: nominal, ordinal, interval and ratio scale.

Attitude Measurement  
and Scaling

## NOTES

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 Attitude is a predisposition of the individual to evaluate some objects or symbol. Attitude has three components: cognitive, affective and intention or action component.

 Scales can be classified as single-item and multiple-item scales. Another classification could be whether the scales are comparative or noncomparative

in nature.

■ The observed measurement need not be equal to the true value of the measurement. Some systematic and random errors may be found in the observed measurement. There are three criteria for determining the accuracy of a measurement—reliability, validity and sensitivity.

## 5.7 KEY WORDS

■ **Balanced scale:** A scale that has equal number of favourable and unfavourable categories.

■ **Comparative scale:** A scale in which respondents make use of some standard frame of reference before answering the question.

■ **Forced scale:** A scale in which the respondent is forced to take a stand

■ **Interval scale:** A scale that makes use of an arbitrary origin.

■ **Validity:** It deals with whether a scale measures what it is supposed to measure.

## 5.8 SELF ASSESSMENT QUESTIONS AND EXERCISES

### Short-Answer Questions

1. What is the meaning of measurement in research? Give examples.
2. Outline the steps involved in constructing a Likert scale.
3. Briefly explain the concepts of reliability, validity and sensitivity.

### Long-Answer Questions

1. Discuss four types of measurements using examples.
2. Define attitude. What are its various components?
3. Explain an itemized rating scale. What are the various issues involved in constructing an itemized rating scale?

## NOTES

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### Attitude Measurement

## PROCESSING AND ANALYSIS OF DATA

Data processing is done for making comparisons and analysis.

Processing implies editing, coding, classification and tabulation of collected data so that they are amenable to analysis.

Analysis refers to computation of certain measures and searching for patterns of relationship that exist among data-groups.

### PROCESSING OPERATIONS

**1. Editing** is done to detect errors and omissions and to correct these when possible.

Purpose of editing is to assure accuracy, consistency with other facts, uniformity, and arranged to facilitate coding and tabulation.

**2. Coding** refers to process of assigning numerals or other symbols to data to facilitate classification /categorization.

**3. Classification:** Large volume of raw data is put into homogeneous groups to draw meaningful relationships.

Classification is done in two ways.

(i) *Classification according to attributes:*

(ii) *Classification according to class-intervals:*

Size of class interval:

$$i = R / (1 + 3.3 \log N)$$

where

$i$  = size of class interval;

$R$  = Range (i.e., difference between Max. – Min.)

$N$  = Number of items to be grouped.

Open-ended intervals in the overall frequency distribution.

Types of class interval

| <i>Exclusive type class intervals:</i> | <i>Inclusive type class intervals</i> |
|----------------------------------------|---------------------------------------|
| 10–20                                  | 11–20                                 |
| 20–30                                  | 21–30                                 |
| 30–40                                  | 31–40                                 |
| 40–50                                  | 41–50                                 |
| For continuous data                    | For discrete data                     |

(iii) Frequency of each class?

This can be done either by tally sheets or by mechanical aids, or data processing software using computer.

### 4. Tabulation:

Tabulation is the process of summarising raw data and displaying the same in compact form in rows and columns for analysis.

Tabulation is essential because

1. It conserves space
2. facilitates the process of comparison
3. facilitates summation of items and detection of errors and omissions

4. provides a basis for various statistical computations

Tabulation may also be classified as simple and complex tabulation.

Simple tabulation gives information about one or more groups of independent questions, whereas complex tabulation shows division of data in two or more categories.

### Complex tabulation

two-way tables -> two inter-related characteristics of data

three-way tables -> three interrelated characteristics of data

..

k-way table (manifold table) -> levels of k categorical variables

### Simple table S(i) -> i represents class

|          | B.Tech | M.Tech | MBA | Total |
|----------|--------|--------|-----|-------|
| Students | 295    | 95     | 55  | 445   |

### Two way table S(i, j) -> i represents class and j represents branch

|             | B.Tech | M.Tech | PhD | Total |
|-------------|--------|--------|-----|-------|
| Civil       | 72     | 30     | 18  | 120   |
| Electrical  | 74     | 34     | 12  | 120   |
| Electronics | 77     | 25     | 15  | 117   |
| Mechanical  | 72     | 36     | 10  | 118   |
| Total       | 295    | 125    | 55  | 475   |

### Three way table S(i, j, k) -> i, j, k represents class, branch and gender

|                | B.Tech |     | M.Tech |    | PhD |    | Total |     |
|----------------|--------|-----|--------|----|-----|----|-------|-----|
|                | M      | F   | M      | F  | M   | F  | M     | F   |
|                | 1      | 2   | 3      | 4  | 5   | 6  |       |     |
| 1. Civil       | 44     | 28  | 12     | 18 | 9   | 9  | 65    | 55  |
| 2. Electrical  | 41     | 33  | 17     | 17 | 7   | 5  | 65    | 55  |
| 3. Electronics | 42     | 35  | 12     | 13 | 7   | 8  | 61    | 56  |
| 4. Mechanical  | 43     | 29  | 20     | 16 | 6   | 4  | 69    | 49  |
| Total          | 170    | 125 | 61     | 64 | 29  | 26 | 260   | 215 |

1. Every table should have a clear, concise and adequate title
2. Every table should be given a distinct number to facilitate easy reference.
3. Column headings (captions) and the row headings (stubs) of table should be clear.
4. Units of measurement under each heading should be indicated.
5. Explanatory footnotes, if any, should be placed directly beneath the table.
6. Source or sources of data must be indicated just below the table.
7. The columns may be numbered to facilitate reference.
8. Those columns whose data are to be compared should be kept side by side.

9. It is generally considered better to approximate figures before tabulation.
10. Miscellaneous and exceptional items, if any, should be placed in the last row of table.

## ELEMENTS/TYPES OF ANALYSIS

Analysis involves

- Estimating the parameters of the population (Descriptive analysis)
- Testing of hypotheses for drawing inferences (Inferential analysis also known as statistical analysis)

**DESCRIPTIVE ANALYSIS** is may be in respect of one variable, or two variables, or more than two variables (described as multivariate analysis).

### Univariate Analysis

#### Population parameters

- (1) measures of central tendency -> Mean, Median, Mode
- (2) measures of dispersion -> Variance, Standard Deviation
- (3) measures of asymmetry (skewness)

### Bivariate Analysis

**Correlation analysis** -> How two variables are related.

**Causal analysis** -> how one or more variables affect changes in another variable. This analysis can be termed as regression analysis.

**Multivariate analysis** -> simultaneously analyse more than two variables

- (a) *Multiple regression analysis*: one dependent variable which is a function of two or more independent variables.
- (b) *Multiple discriminant analysis*: used for categorical DV on basis of some attribute.
- (c) *Multivariate analysis of variance (or multi-ANOVA)*:

**INFERENTIAL ANALYSIS** is concerned with testing hypotheses for desired level of significance.

*Inferential statistics* are also known as **sampling statistics**

Testing of Hypothesis

|                        |                                                                             |
|------------------------|-----------------------------------------------------------------------------|
| 1. z test<br>2. t test | 3. Chi square test<br>4. Sign Test<br>5. Run Test<br>6. Kruskal-Wallis test |
|------------------------|-----------------------------------------------------------------------------|

## What Is Hypothesis Testing in Statistics?

Hypothesis Testing is a type of statistical analysis in which you put your assumptions about a population parameter to the test. It is used to estimate the relationship between 2 statistical variables.

### Null Hypothesis and Alternative Hypothesis

The Null Hypothesis is the assumption that the event will not occur. A null hypothesis has no bearing on the study's outcome unless it is rejected.

The Alternate Hypothesis is the assumption that is to be proved. The acceptance of the alternative hypothesis follows the rejection of the null hypothesis.  $H_1$  is the symbol for it.

### Hypothesis Testing Formula

$$Z = (\bar{x} - \mu_0) / (\sigma / \sqrt{n})$$

- Here,  $\bar{x}$  is the sample mean,
- $\mu_0$  is the population mean,
- $\sigma$  is the standard deviation,
- $n$  is the sample size.

### How Hypothesis Testing Works?

An analyst performs hypothesis testing on a statistical sample. Measurements and analyses are conducted on a random sample of the population to test two hypotheses: the null and alternative hypotheses.

### Steps in Hypothesis Testing

#### Formulate Hypotheses

- Null Hypothesis ( $H_0$ ): This hypothesis states that there is no effect or difference, and it is the hypothesis you attempt to reject with your test.
- Alternative Hypothesis ( $H_1$ ): This hypothesis is what you might believe to be true or hope to prove true. It is usually considered the opposite of the null hypothesis.

#### Choose the Significance Level ( $\alpha$ )

The significance level, often denoted by **alpha ( $\alpha$ )**, is the probability of rejecting the null hypothesis when it is true. Common choices for  $\alpha$  are 0.05 (5%), 0.01 (1%), and 0.10 (10%).

#### Select the Appropriate Test

Choose a statistical test based on the type of data and the hypothesis. Common tests include **t-tests, chi-square tests, ANOVA, and regression analysis**. The selection depends on data type, distribution, sample size, and whether the hypothesis is one-tailed or two-tailed.

#### Collect Data

Gather the data that will be analyzed in the test. To infer conclusions accurately, this data should be representative of the population.

#### Calculate the Test Statistic

Based on the collected data and the chosen test, calculate a test statistic that reflects how much the observed data deviates from the null hypothesis.

## Determine the p-value

The p-value is the probability of observing test results at least as extreme as the results observed, assuming the null hypothesis is correct. It helps determine the strength of the evidence against the null hypothesis.

## Make a Decision

Compare the p-value to the chosen significance level:

- If the  $p\text{-value} \leq \alpha$ : Reject the null hypothesis, suggesting sufficient evidence in the data supports the alternative hypothesis.
- If the  $p\text{-value} > \alpha$ : Do not reject the null hypothesis, suggesting insufficient evidence to support the alternative hypothesis.

## Types of Hypothesis Testing

### 1. Z Test

To determine whether a discovery or relationship is statistically significant, hypothesis testing uses a z-test. It usually checks to see if two means are the same (the null hypothesis). Only when the population standard deviation is known and the sample size is 30 data points or more, can a z-test be applied.

### 2. T Test

The “t-test” or “Student’s t-test” developed by William Sealy Gosset, (who wrote under the pseudonym “Student”) compares the means of two groups to see if they are significantly different from each other.

There are different types of t-tests, such as:

- Independent t-test: Compares the means of two independent groups.
- Paired t-test: Compares means from the same group at different times.
- One-sample t-test: Compares the mean of a single group against a known mean.

It is frequently used in hypothesis testing.

### 3. Chi-Square

Chi-square test is used for hypothesis testing to determine if the expected and observed results are well-fitted. The Chi-square test analyzes the differences between categorical variables from a random sample. The test's fundamental premise is that the observed values in the data should be compared to the predicted values that would be present if the null hypothesis were true.

### 4. ANOVA

ANOVA, or Analysis of Variance, is a statistical method used to compare **two or more groups**. It's particularly useful when there are significant differences between multiple groups. For instance, in business, a company might use ANOVA to analyze whether three different stores are performing differently in terms of sales. It's widely used in fields like medical research and social sciences, where comparing group differences can provide valuable insights.

## Type 1 and Type 2 Error

A hypothesis test can result in two types of errors.

Type 1 Error: A Type-I error occurs when sample results reject the null hypothesis despite being true.



Type 2 Error: A Type-II error occurs when the null hypothesis is not rejected when it is false, unlike a Type-I error.

### Example Problems on Hypothesis Testing

Here are the practice problems on hypothesis testing that will help you understand how to apply these concepts in real-world scenarios:

#### Example:

The mean marks scored by students in an examination was 67.39 with standard deviation 1.30. Then students were taught using a new teaching methodology. After being taught in new teaching methodology students appeared in a similar examination. The mean marks scored by a sample of 60 students was found to be 67.47. Analyse the usefulness of new teaching method.

#### Solution:

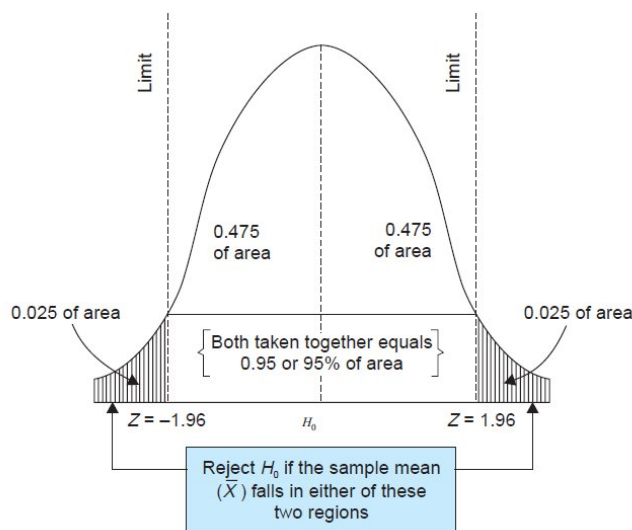
$H_0$ : There is no effect of new teaching method

$H_a$ : The new teaching method is effective in improving the marks of students.

Given information:  $\mu = 67.39$   $\sigma_p = 1.30$ ,  $\bar{X} = 67.57$ ,  $n = 60$ .

$$z = \frac{\bar{X} - \mu}{\sigma_p / \sqrt{n}} = (67.57 - 67.39) / (1.30 / \sqrt{60}) = 1.07$$

P value corresponding to  $z=1.07$  is 0.142

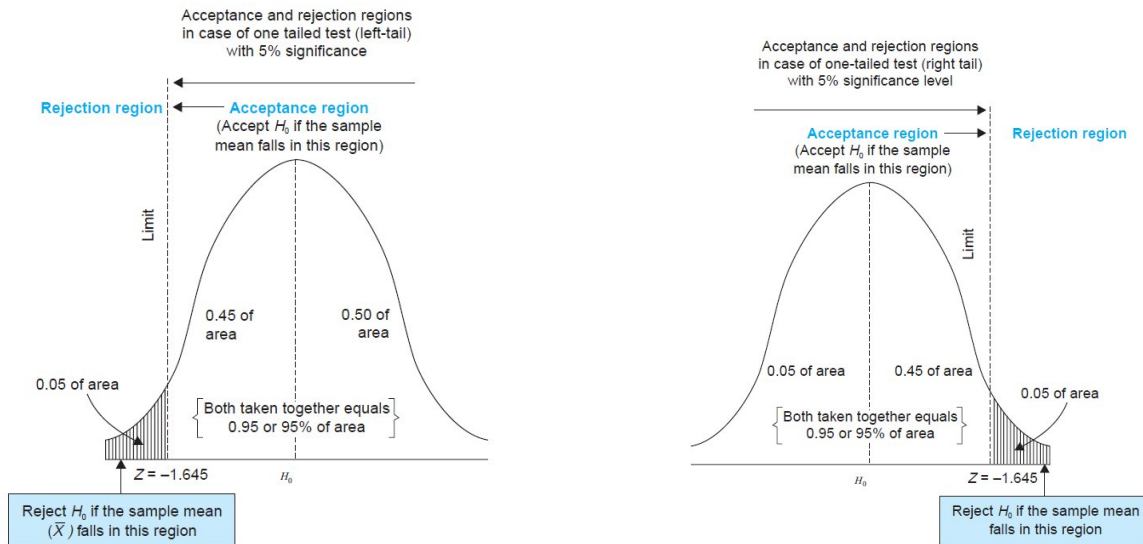


Since

$-1.96 < 1.07 < 1.96$  i.e. p value  $0.142 > 0.05$  the null hypothesis is not rejected.

$H_0$ : The new teaching method is not effective in improving the marks of students.

$H_a$ : The new teaching method is effective in improving the marks of students.



Since  $1.07 < 1.645$  i.e. p value  $0.142 > 0.05$  the null hypothesis is not rejected.

By not rejecting the null hypothesis, we might be committing type-II error. It is possible that the new teaching method is actually effective in improving marks of students.

Prob. Corresponding to  $z = 1.07$  is 0.8577

Prob (improvement) – 14.23%

Type-II error  $\beta = 0.1423$

### Example:

The mean marks scored by students in an examination was 67.39 with standard deviation 1.30. Then students were taught using a new teaching methodology. After being taught in new teaching methodology students appeared in a similar examination. The mean marks scored by a sample of 50 students was found to be 67.90. Analyse the usefulness of new teaching method.

$$z = \frac{\bar{X} - \mu}{\sigma_p / \sqrt{n}} = (67.90 - 67.39) / (1.30 / \sqrt{50}) = 2.77$$

p value corresponding to  $z = 2.77$  is 0.0028

Since  $2.77 > 1.645$  ie p value  $0.0028 < 0.05$  the null hypothesis is rejected.

$H_0$ : The new teaching method is not effective in improving the marks of students.

By rejecting the null hypothesis, we might be committing type-I error. It is possible that the new teaching method is actually not effective in improving marks of students.

Type-I error  $\alpha = 0.0028$

Power of test is  $(1 - \beta) = 1 - 0.0028 = 99.77\%$

- Definition of t test: “ It’s a method of testing hypothesis about the mean of small sample drawn from a normally distributed population when the standard deviation for the sample is unknown.”

Both t-tests and z-tests are used to compare means and test hypotheses, but they have their unique advantages and limitations:

### **t-Test**

Advantages:

1. Small Sample Size: Ideal for small sample sizes (typically  $n < 30$ ).
2. Unknown Population Standard Deviation: Used when the population standard deviation is unknown and estimated from the sample.
3. Flexibility: Can be used for paired samples, independent samples, and one-sample tests.

Limitations:

1. Less Robust with Large Samples: Although it can be used, it becomes less efficient compared to a z-test as sample sizes increase.
2. Normality Assumption: Assumes the sample data is approximately normally distributed, which might not hold true for very small samples.

### **z-Test**

Advantages:

1. Large Sample Size: Ideal for large sample sizes (typically  $n > 30$ ).
2. Known Population Standard Deviation: More efficient when the population standard deviation is known.
3. Simplicity: Computationally simpler when conditions for its use are met.

Limitations:

1. Not Suitable for Small Samples: Less reliable for small sample sizes.
2. Requires Known Population Standard Deviation: Not applicable when the population standard deviation is unknown and has to be estimated from the sample.

In summary, use a t-test when dealing with small sample sizes or unknown population standard deviations, and a z-test for large sample sizes with known population standard deviations. Clear which one you need to use for your data?

### **Assumptions of t-Test**

Dependent variables are interval or ratio.

The population from which samples are drawn is normally distributed.

Samples are randomly selected.

The groups have equal variance (Homogeneity of variance).

The t-statistic is robust (it is reasonably reliable even if assumptions are not fully met.)

### **Applications of t test**

The calculation of a confidence interval for a sample mean.

To test whether a sample mean is different from a hypothesized value.

To compare mean of two samples.

To compare two sample means by group.

## Types of "t" test

**Single sample t-test** — we have only 1 group; want to test against a hypothetical mean.

**Independent samples t-test** — we have 2 means, 2 groups; no relation between groups, Eg: When we want to compare the mean of T/T group with Placebo group.

**Paired t-test** — It consists of samples of matched pairs of similar units or one group of units tested twice. Eg: Difference of mean pre & post drug intervention.

## Single Sample t-test

It is used in measuring whether a sample value significantly differs from a hypothesized value. For example, a research scholar might hypothesize that on an average it takes 3 minutes for people to drink a standard cup of coffee.

He conducts an experiment and measures how long it takes his subjects to drink a standard cup of coffee.

The one sample t-test measures whether the mean amount of time it took the experimental group to complete the task varies significantly from the hypothesized 3 minutes value.

## Equation for a one-sample t-test

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

where

$t$  = the  $t$  statistic

$\bar{x}$  = the mean of the sample

$\mu$  = the comparison mean

$s$  = the sample standard deviation

$n$  = the sample size

## Example

67, 67, 67, 68, 66, 67, 66, 67, 67, 70, 69, 66, 66, 67, 68, 67, 67, 68, 68, 71

Mean  $\bar{X}$  = 67.45  $\sigma_s$  = 1.32

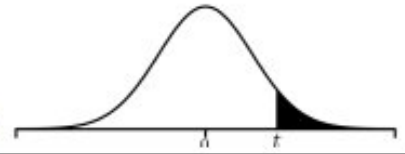
Is the mean different from population mean  $\mu$  = 67.39

$$t = \frac{67.45 - 67.39}{1.32/\sqrt{20}} = 0.20$$

For 19 df critical value of  $t$  for significance level 0.05 is 1.73

Since  $0.20 < 1.73$ , The score has not improved is **not rejected**.

## Critical Values for Student's *t*-Distribution



| df | Upper Tail Probability: $\Pr(T > t)$ |       |        |        | df | Upper Tail Probability: $\Pr(T > t)$ |       |       |       |
|----|--------------------------------------|-------|--------|--------|----|--------------------------------------|-------|-------|-------|
|    | 0.1                                  | 0.05  | 0.02   | 0.01   |    | 0.1                                  | 0.05  | 0.02  | 0.01  |
| 1  | 3.078                                | 6.314 | 15.895 | 31.821 | 11 | 1.363                                | 1.796 | 2.328 | 2.718 |
| 2  | 1.886                                | 2.920 | 4.849  | 6.965  | 12 | 1.356                                | 1.782 | 2.303 | 2.681 |
| 3  | 1.638                                | 2.353 | 3.482  | 4.541  | 13 | 1.350                                | 1.771 | 2.282 | 2.650 |
| 4  | 1.533                                | 2.132 | 2.999  | 3.747  | 14 | 1.345                                | 1.761 | 2.264 | 2.624 |
| 5  | 1.476                                | 2.015 | 2.757  | 3.365  | 15 | 1.341                                | 1.753 | 2.249 | 2.602 |
| 6  | 1.440                                | 1.943 | 2.612  | 3.143  | 16 | 1.337                                | 1.746 | 2.235 | 2.583 |
| 7  | 1.415                                | 1.895 | 2.517  | 2.998  | 17 | 1.333                                | 1.740 | 2.224 | 2.567 |
| 8  | 1.397                                | 1.860 | 2.449  | 2.896  | 18 | 1.330                                | 1.734 | 2.214 | 2.552 |
| 9  | 1.383                                | 1.833 | 2.398  | 2.821  | 19 | 1.328                                | 1.729 | 2.205 | 2.539 |
| 10 | 1.372                                | 1.812 | 2.359  | 2.764  | 20 | 1.325                                | 1.725 | 2.197 | 2.528 |

### Example 2:

70      69      66      69      70      67      69      68      70      69      68

68.64

1.29

Is the mean different from population mean  $\mu = 67.39$

$$t = \frac{68.64 - 67.39}{1.29/\sqrt{10}} = 3.06$$

For 9 df critical value of *t* for significance level 0.05 is 1.83

Since  $3.06 > 1.73$ , The score has not improved is **rejected**.

## Independent samples *t*-test

It is also called unpaired *t*-test/ two sample *t* test.

It is the *t*-test to be used when two separate independent and identically distributed variables are measured.

$$t = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Eg:

1. Comparison of quality of life improved for patients who took drug Valporate as opposed to patients who took drug Levetiracetam in myoclonic seizures.
2. Comparison of mean cholesterol levels in treatment group with placebo group after administration of test drug.

78, 82, 87, 65, 75, 82, 71       $\bar{x}_1 = 77.14$      $\sigma_1 = 7.47$

87, 82, 86, 95, 73                       $\bar{x}_2 = 84.60$      $\sigma_2 = 8.02$

Calculate t-value =  $7.46 / 4.565 = 1.633$  (p value is coming to 0.07)

For the .05 probability level with 10 degrees of freedom,  
the critical t score is 2.228 for a two-tailed test and 1.812 for a one-tailed test.

Since the calculated t score is lower than the critical t-score, improvement if any is not significant at the .05 probability level. Cd

### Paired t test

Eg: I. pre-test/post-test samples I n which a factor is measured before and after an intervention,

° Suppose a sample of “n” subjects were given an antihypertensive drug we want to check blood pressure before and after treatment. We want to find out the effectiveness of the treatment by comparing mean pre & post t/t.

' To test the null hypothesis that the true mean difference is zero, the procedure is as follows:

1. Calculate the difference ( $d_i = y_i - x_i$ ) between the two observations on each pair.
2. Calculate the mean difference, d and standard deviation  $\sigma_d$
3. Calculate the standard error of the mean differences.  $e = \sigma_d / \sqrt{n}$
4. Calculate the t-statistic, which is given by  $T = d/e$ , Under the null hypothesis, this statistic follows a t-distribution with  $n - 1$  degrees of freedom.
5. Use tables of the t-distribution to compare your value for T to the t distribution.

|       |    |    |    |    |    |    |    |       |       |       |       |
|-------|----|----|----|----|----|----|----|-------|-------|-------|-------|
| $x_i$ | 78 | 82 | 87 | 65 | 75 | 82 | 71 |       |       |       |       |
| $y_i$ | 82 | 84 | 86 | 72 | 81 | 85 | 72 | d     | SD    | SE    | T     |
| $d_i$ | 4  | 2  | -1 | 7  | 6  | 3  | 1  | 3.143 | 2.795 | 1.056 | 2.976 |

t-value at 0.05 significance level is 1.833

Since  $2.976 > 1.833$ , or corresponding p-value  $0.012 < 0.05$ , null hypothesis is rejected.

A **Non-Parametric Hypothesis Test**, also known as a distribution-free method, is a statistical procedure that assesses whether a hypothesis is true or false without making assumptions about the distribution of the data. Non-parametric tests are used when the assumptions of a parametric test are not met. Can adapt to ordinal and categorical data.

Some non-parametric tests are: Chi-square test, Sign test, Run test, Kruskal-Wallis test, Mann-Whitney-Wilcoxon test, and Kolmogorov-Smirnov test.

A chi-square test is a statistical hypothesis test that determines if data is significantly different from what is expected. It compares **observed values** in data to the **expected values if the null hypothesis is true**.

- SPSS Tutorials: Chi-Square Test of Independence - LibGuides

10 Jul 2024 — This test utilizes a contingency table to analyze the data. A contingency table (also known as a cross-tabulation, cros...

Kent State University

Example:

A University conducted a survey on health awareness among its students. The survey revealed that only a minor proportion of 15% of students do physical exercise/ yoga regularly, whereas 25% do physical exercise/ yoga sporadically, and rest 60% do not do any exercise/ yoga.

The University launched a health promotion campaign on campus. To evaluate the impact of the program, the University again conducted a survey.

Out of 470 students surveyed, ed graduates and asked the same questions. The survey was completed by 470 graduates and the following data as obtained.

|             | No<br>Exercise | Sporadic<br>Exercise | Regular<br>Exercise | Total |
|-------------|----------------|----------------------|---------------------|-------|
| Observation | 255            | 125                  | 90                  | 470   |

Has the health promotion campaign caused any change in behaviour of students towards exercise/yoga?

### Hypotheses and level of significance.

Null Hypothesis  $H_0$  – There is no change.

Alt. Hypothesis  $H_a$  -- Null hypothesis is false.

Predict the outcome of survey at 5% level of significance.

|             | No<br>Exercise | Sporadic<br>Exercise | Regular<br>Exercise | Total |
|-------------|----------------|----------------------|---------------------|-------|
| Observation | 255            | 125                  | 90                  | 470   |
| Expected    | 60%            | 25%                  | 15%                 |       |
|             | 282.0          | 117.5                | 70.5                |       |

The test statistic is:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$$\chi^2 = \frac{(255-282)^2}{282} + \frac{(125-117.5)^2}{117.5} + \frac{(90-70.5)^2}{70.5}$$

$$\chi^2 = 2.59 + 0.48 + 5.39 = 8.46$$

Decision Rule to accept or reject Null hypothesis will depend on level of significance and the degrees of freedom.

Critical values can be found in a table of probabilities for the  $\chi^2$  distribution. Here we have  $df=k-1=3-1=2$  and a 5% level of significance. The appropriate critical value is 5.99, and the decision rule is as follows: Reject  $H_0$  if  $\chi^2 \geq 5.99$ .

### Example 2:

Data on the distribution of weight (in categories) based on BMI index among people in 2002 were as follows: 2% Underweight, 39% Normal Weight, 36% Overweight, and 23% Obese. Suppose we want to assess whether over the years the distribution of BMI has changed. Using data from the  $n=3,326$  participants observed the following:



| Under weight BMI<br><18.5 | Normal Weight BMI<br>18.5-24.9 | Over weight BMI<br>25.0-29.9 | Obese BMI<br>≥ 30 | Total |
|---------------------------|--------------------------------|------------------------------|-------------------|-------|
| 20                        | 932                            | 1374                         | 1000              | 3326  |

$H_0$ : Distribution of responses is 0.02, 0.39, 0.36, 0.23

$H_1$ :  $H_0$  is false.  $\alpha=0.05$

|   | Under weight BMI<br><18.5 | Normal Weight BMI<br>18.5-24.9 | Over weight BMI<br>25.0-29.9 | Obese BMI<br>≥ 30 | Total |
|---|---------------------------|--------------------------------|------------------------------|-------------------|-------|
| O | 20                        | 932                            | 1374                         | 1000              | 3326  |
| E | 66.5                      | 1297.1                         | 1107.4                       | 765               |       |

$$\chi^2 = \frac{(20-66.5)^2}{66.5} + \frac{(932-1297.4)^2}{1297.4} + \frac{(1374-1197.4)^2}{1197.4} + \frac{(1000-765.0)^2}{765.0}$$

$$\chi^2 = 32.52 + 102.77 + 26.05 + 72.19 = 233.53$$

Decision rule.

df=k-1=4-1=3 and 5% level of significance ( $\alpha=0.05$ ). The appropriate critical value is 7.81

Since  $\chi^2 > 7.81$ , Reject  $H_0$

### Advantages of the Chi-Square Test:

|                     |                        |
|---------------------|------------------------|
| Flexibility         | Non-parametric Nature: |
| Ease of Calculation | Versatility            |
| Categorical data    | Large Sample Sizes:    |

### Limitations of the Chi-Square Test:

|                                     |                       |
|-------------------------------------|-----------------------|
| Sample Size Sensitivity Requirement | Expected Frequency    |
| Independence Assumption             | Non-parametric Nature |
| Categorical Data Limitation         |                       |

Doesn't provide information about strength of the relationship or its nature

## Two types of Chi-Square Tests

### 1. Chi-Square Goodness-of-Fit Test

- To determine if a sample data matches a population with a specific distribution.

### 2. Chi-Square Test of Independence

- To determine if there is a significant association between categorical variables. e.g. Checking if gender and preference for a new product are related.

#### Example:

|                   | No Regular Exercise | Sporadic Exercise | Regular Exercise | Total |
|-------------------|---------------------|-------------------|------------------|-------|
| Out Campus Hostel | 32                  | 30                | 28               | 90    |
| In Campus Hostel  | 74                  | 64                | 42               | 180   |
| Home Apartments   | 110                 | 25                | 15               | 150   |
| Home Single House | 39                  | 6                 | 5                | 50    |
| Total             | 255                 | 125               | 90               | 470   |

Hypotheses and level of significance.

$H_0$ : Living arrangement has no effect on exercise

$H_1$ :  $H_0$  is false.  $\alpha=0.05$

|                   | No Regular Exercise | Sporadic Exercise | Regular Exercise | Total |
|-------------------|---------------------|-------------------|------------------|-------|
| Out Campus Hostel | 32<br>(48.8)        | 30<br>(23.9)      | 28<br>(17.2)     | 90    |
| In Campus Hostel  | 74<br>(97.7)        | 64<br>(47.9)      | 42<br>(34.5)     | 180   |
| Home Apartments   | 110                 | 25                | 15               | 150   |

|                   |              |             |            |     |
|-------------------|--------------|-------------|------------|-----|
|                   | (81.4)       | (39.9)      | (28.7)     |     |
| Home Single House | 39<br>(27.1) | 6<br>(13.3) | 5<br>(9.6) | 50  |
| Total             | 255          | 125         | 90         | 470 |

$$\chi^2 = \frac{(32-48.8)^2}{48.8} + \frac{(30-23.9)^2}{23.9} + \frac{(28-17.2)^2}{17.2} \\ + \frac{(74-48.8)^2}{48.8} + \frac{(64-47.9)^2}{47.9} + \frac{(42-34.5)^2}{34.5} \\ + \frac{(110-81.4)^2}{81.4} + \frac{(25-39.9)^2}{39.9} + \frac{(15-28.7)^2}{28.7} \\ + \frac{(39-27.1)^2}{27.1} + \frac{(6-13.3)^2}{13.3} + \frac{(5-9.6)^2}{9.6}$$

$$\chi^2 = 5.78 + 1.58 + 6.78 + 5.75 + 5.41 + 1.63 \\ + 10.05 + 5.58 + 6.54 + 5.23 + 4.01 + 2.20 = 60.5$$

For two way data table

df = (r-1)(c-1), where r and c are numbers of rows and columns.

df = (4-1)(3-1) = 6.

For df=6 and 5% level of significance ( $\alpha = 0.05$ ),

Critical value  $\chi^2$  is 12.59.

Since  $60.5 \geq 12.59$ , Reject  $H_0$

Prob:

|                   |                     |                   |                  |       |
|-------------------|---------------------|-------------------|------------------|-------|
|                   | No Regular Exercise | Sporadic Exercise | Regular Exercise | Total |
| Out Campus Hostel | 32                  | 30                | 28               | 90    |
| In Campus Hostel  | 74                  | 64                | 42               | 180   |

Is there any difference between Out Campus Hostel and In Campus Hostel?

## Sign Test

It is a simple nonparametric test for matched or paired data.

| Child | Before Treatment | After 1 Week of Treatment | Difference (Before-After) | Sign |
|-------|------------------|---------------------------|---------------------------|------|
| 1     | 85               | 75                        | 10                        | +    |
| 2     | 70               | 50                        | 20                        | +    |
| 3     | 40               | 50                        | -10                       | -    |
| 4     | 65               | 40                        | 25                        | +    |
| 5     | 80               | 20                        | 60                        | +    |
| 6     | 75               | 65                        | 10                        | +    |
| 7     | 55               | 40                        | 15                        | +    |
| 8     | 20               | 25                        | -5                        | -    |

$H_0$ : There is no difference due to treatment

$H_1$ :  $H_0$  is false.  $\alpha=0.05$

The approach is to analyze only the signs of the difference scores

Test statistic is number of positive signs or number of negative signs, whichever is smaller.  $C = \min(2, 6) = 2$

### Critical value

For Sample size, number of matched pairs i.e.  $n=8$   
and level of significance  $\alpha=0.05$ .

Critical value is 1 (From Sign Test Table)

Since  $2 > 1$ , Do not reject  $H_0$

The Critical Value can be calculated using Binomial Theorem

Example:

| S.No. | Control | Treatment | Sign |
|-------|---------|-----------|------|
|-------|---------|-----------|------|

|   |      |      |   |
|---|------|------|---|
| 1 | 7.95 | 10.3 | + |
| 2 | 7.05 | 7.97 | + |
| 3 | 4.73 | 4.83 | + |
| 4 | 4.96 | 5.51 | + |

$H_0$  : There is no improvement

Test with 5% level of significance

If null hypothesis is true, probability of improvement  $p=0.5$  and no improvement  $q = 1- p = 0.5$

In 4 trials (samples), all are success.

Prob of 4 successes in 4 trials

$$P(x \text{ successes}) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}$$

Here  $n = 4$ ,  $x = 4$ , and  $p = 0.5$

So  $P = 1 \times (0.5)^4 \times (0.5)^0 = 0.0625$

Since  $P > 0.05$ , The Null hypothesis is not rejected. We need more number of trials /samples.

If there is no effect

Prob. of no Improvement  $p = 0.5$  and improvement  $q = (1- 0.5) = 0.5$

$$P(x \text{ successes}) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}$$

Probability of success in 8 trials

$n = 8$ ,  $p = 0.5$ ,  $x = 0, 1, 2, \dots 7, 8$

|   |        |   |        |   |        |
|---|--------|---|--------|---|--------|
| 0 | 0.0039 | 3 | 0.2188 | 6 | 0.1094 |
| 1 | 0.0313 | 4 | 0.2734 | 7 | 0.0313 |
| 2 | 0.1094 | 5 | 0.2188 | 8 | 0.0039 |

| Number of Improvement (n) | Prob. P(n) | Cum. Prob. P(x≤n) | Prob. of Improvement | Prob. of no Improvement P(x>n) |
|---------------------------|------------|-------------------|----------------------|--------------------------------|
| 0                         | 0.004      | 0.004             | 0.004                | 0.996                          |
| 1                         | 0.031      | 0.035             | 0.035                | 0.965                          |

|   |       |       |       |                     |
|---|-------|-------|-------|---------------------|
| 2 | 0.109 | 0.145 | 0.145 | 0.855               |
| 3 | 0.219 | 0.363 | 0.363 | 0.637               |
| 4 | 0.273 | 0.637 | 0.637 | 0.363               |
| 5 | 0.219 | 0.856 | 0.856 | 0.144               |
| 6 | 0.109 | 0.965 | 0.965 | <b><u>0.034</u></b> |
| 7 | 0.031 | 0.996 | 0.996 | 0.004               |
| 8 | 0.004 | 1     | 1     | 0.0                 |

| Number of Sign Change | Prob. | Prob. of Improvement           | Prob. of no difference |
|-----------------------|-------|--------------------------------|------------------------|
| 0                     | 0.008 | $P(x=0)$ or $P(x=8)$           | 0.008                  |
| 1                     | 0.062 | $P(x \leq 1)$ or $P(x \geq 7)$ | 0.07                   |
| 2                     | 0.218 | $P(x \leq 2)$ or $P(x \geq 6)$ | 0.288                  |
| 3                     | 0.438 | $P(x \leq 3)$ or $P(x \geq 5)$ | 0.726                  |
| 4                     | 0.273 | $P(x \leq 4)$ or $P(x \geq 4)$ | 1.0                    |

$$P(x \leq 1) = P(0) + P(1) = 0.008 + 0.031 = 0.035$$

### Limitations of the Sign Test:

1. **Loss of Information:** Considers the direction of changes (positive or negative). This can result in a loss of valuable information, if the magnitude is important to the analysis.
2. **Low Power:** Less powerful compared to parametric tests (like the paired t-test). Requires a larger sample size for same level of significance.
3. **Ties Handling:** In cases where there are a lot of ties (no change between pairs), the sign test may not be effective.
4. **Simple Hypotheses Only:** It does not allow for more complex hypotheses or comparisons involving variances or distributions.

**Applicability:** Best used when data does not meet the assumptions for parametric tests, such as normality or equal variances. For datasets

that meet these assumptions, parametric tests are generally preferred for their greater power and sensitivity.

### One sided vs two sided test

|                                                                                                                                                                                         |                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>If '+' indicates improvement<br/> <math>H_0</math> : No improvement is rejected at <math>\alpha = 0.05</math><br/>         If <math>x \geq N</math> <math>H_0</math> is rejected</p> | <p><math>H_0</math> : No Difference is rejected at <math>\alpha = 0.05</math><br/>         If <math>x \geq N_1</math> and <math>x \leq N_2</math> the <math>H_0</math> is rejected</p>                    |
| <p>If <math>x \geq 7</math><br/> <math>H_0</math> : No improvement is rejected at <math>\alpha = 0.05</math></p>                                                                        | <p>If <math>(x \leq 1)</math> or <math>P(x \geq 7)</math><br/> <math>0.035 + 0.035 = 0.07</math><br/> <math>H_0</math> : No difference <b><u>cannot be rejected</u></b> at <math>\alpha = 0.05</math></p> |

### RUN Test

The runs test is a statistical test to determine whether the data is random or whether there is a pattern.

The runs test is a type of non-parametric test and hence, there is no need for the assumption of a normal distribution to hold true.

Example:

*F F M M F M F F F F M F F M M M M M F F*

Runs :  $\underbrace{F F}_1 \underbrace{M M}_2 \underbrace{F}_3 \underbrace{M}_4 \underbrace{F F F F}_5 \underbrace{M}_6 \underbrace{F F}_7$

Runs :  $\underbrace{M M M M M}_8 \underbrace{F F}_9$

Total  $n = 20$       Female ( $n_1$ ) = 11,      Male ( $n_2$ ) = 9      runs ( $r$ ) = 9.

The normal distribution via the following formula:

$$z = (r - \mu_r) / \sigma_r$$

where:

- $r$  is the actual number of runs;
- $\mu_r$  is the expected number of runs; and
- $\sigma_r$  is the standard deviation of the number of runs.

The values of  $\mu_r$  and  $\sigma_r$  are computed as follows:

$$\mu_r = \frac{2n_1n_2}{n_1 + n_2} + 1$$

$$\sigma_r = \sqrt{\frac{(2n_1n_2)(2n_1n_2 - n_1 - n_2)}{(n_1 + n_2)^2 (n_1 + n_2 - 1)}}$$

Let's calculate for mean and standard deviation of the runs:

$$\mu_r = \frac{2n_1n_2}{n_1+n_2} + 1 \quad \mu_r = \frac{2 * 11 * 9}{11 + 9} + 1 = 10.9$$

And

$$\sigma_r = \sqrt{\left[ \frac{(2 * 11 * 9)(2 * 11 * 9 - 11 - 9)}{(11 + 9)^2 (11 + 9 - 1)} \right]} = 2.15$$

$$z = (r - \mu_r) / \sigma_r = (9 - 10.9) / 2.15 = -0.884$$

At the 5% significance level, a test statistic with an absolute value greater than 1.96 indicates non-randomness (reject  $H_0$ ).

Example: Blood test samples. The data-set are as follow:

7.7, 10.0, 8.8, 12.1, 7.4, 6.1, 6.1, 6.1, 8.0, 8.8,  
6.1, 10.9, 5.0, 12.6, 11.7, 11.5, 6.4, 8.9, 7.2, 7.2,  
6.5, 10.3, 7.9, 7.8, 14.0, 6.8, 7.7, 7.8, 6.7, 7.7

Numerical data is converted to categorical data by grouping using a cut-off point. The cut-off points can be mean, median, mode or by custom. By custom, let us take 7 as cutoff point to categorise data into two groups, good (G) or poor (P).

The data are grouped as follows:

PPPPP GGG PP G P G PPP G PPP G  
PPPP G PP G P

$$n_1 = 21 \quad n_2 = 9$$

Using equations for  $\mu_r$  and  $\sigma_r$

We get  $\mu_r = 13.6$  and  $\sigma_r = 2.245$



Number of runs  $r = 15$

So  $z = (r - \mu_r) / \sigma_r = (15 - 13.6) / 2.245 = 0.624$

At the 5% significance level, a test statistic with an absolute value greater than 1.96 indicates non-randomness.

Since test statistics  $< 1.96$ , the data is random, i.e. No pattern is detected

### Kruskal-Wallis Test

The Kruskal Wallis test is used to compare **medians among k comparison groups** ( $k > 2$ ) and is sometimes described as an ANOVA with the data replaced by their ranks.

The procedure for the test involves pooling the observations from the  $k$  samples into one combined sample in order of rank.

Example: A clinical study is designed to assess differences in albumin levels in adults following diets with different amounts of protein. Three diets are compared, ranging from 5% to 15% protein. The albumin levels of participants following each diet are shown below.

| 5% Protein | 10% Protein | 15% Protein |
|------------|-------------|-------------|
| 3.1        | 3.8         | 4.0         |
| 2.6        | 4.1         | 5.5         |
| 2.9        | 2.9         | 5.0         |
|            | 3.4         | 4.8         |
|            | 4.2         |             |

Is there is a **significant** difference in serum albumin levels among subjects on the three different diets.

Sample sizes are small and not equal across comparison groups ( $n_1=3$ ,  $n_2=5$ ,  $n_3=4$ ).

$H_0$ : There is no difference between three population medians

To conduct the test, we first order the data in the combined total sample of 12 subjects from smallest to largest. We also need to keep track of the group assignments in the total sample.

|  |  |  | Total Sample<br>(Ordered Smallest to<br>Largest) | Ranks |
|--|--|--|--------------------------------------------------|-------|
|--|--|--|--------------------------------------------------|-------|

| T1: 5% Protein | T2: 10% Protein | T3: 15% Protein | 5% Protein | 10% Protein | 15% Protein | 5% Protein | 10% Protein | 15% Protein |
|----------------|-----------------|-----------------|------------|-------------|-------------|------------|-------------|-------------|
| 3.1            | 3.8             | 4.0             | 2.6        |             |             | 1          |             |             |
| 2.6            | 4.1             | 5.5             | 2.9        | 2.9         |             | 2.5        | 2.5         |             |
| 2.9            | 2.9             | 5.0             | 3.1        |             |             | 4          |             |             |
|                | 3.4             | 4.8             |            | 3.4         |             |            | 5           |             |
|                | 4.2             |                 |            | 3.8         |             |            | 6           |             |
|                |                 |                 |            |             | 4.0         |            |             | 7           |
|                |                 |                 |            | 4.1         |             |            | 8           |             |
|                |                 |                 |            | 4.2         |             |            | 9           |             |
|                |                 |                 |            |             | 4.8         |            |             | 10          |
|                |                 |                 |            |             | 5.0         |            |             | 11          |
|                |                 |                 |            |             | 5.5         |            |             | 12          |
|                |                 |                 |            |             |             | 7.5        | 30.5        | 40          |

### Test Statistic for the Kruskal Wallis Test

The test statistic for the Kruskal Wallis test is denoted H and is defined as follows:

$$H = \left( \frac{12}{N(N+1)} \sum_{j=1}^k \frac{R_j^2}{n_j} \right) - 3(N+1)$$

where k=the number of comparison groups, N= the total sample size,  $n_j$  is the sample size in the  $j^{\text{th}}$  group and  $R_j$  is the sum of the ranks in the  $j^{\text{th}}$  group.

In this example  $R_1 = 7.5$ ,  $R_2 = 30.5$ , and  $R_3 = 40$ .

The H statistic for this example is computed as follows: .

$$H = \left( \frac{12}{N(N+1)} \sum_{j=1}^k \frac{R_j^2}{n_j} \right) - 3(N+1) = \frac{12}{12(13)} \left( \frac{7.5^2}{3} + \frac{30.5^2}{5} + \frac{40^2}{4} \right) - 3(13) = 7.52$$

Compare with critical value of H.

If  $H \geq$  critical value, reject  $H_0$  in favour of  $H_1$

From **Kruskal Wallis Test table** the critical value is 5.656.

Since  $7.52 \geq 5.656$ , Reject  $H_0$

Example:

A personal trainer is interested in comparing the anaerobic thresholds of athletes. Anaerobic threshold is important parameter for upper limit of aerobic exercise. The following data are anaerobic thresholds for distance runners, distance cyclists, distance swimmers and cross-country skiers.

| Distance Runners (A) | Distance Cyclists (B) | Distance Swimmers (C) | Cross-Country Skiers (D) |
|----------------------|-----------------------|-----------------------|--------------------------|
| 185                  | 190                   | 166                   | 201                      |
| 179                  | 209                   | 159                   | 195                      |
| 192                  | 182                   | 170                   | 180                      |
| 165                  | 178                   | 183                   | 187                      |
| 174                  | 181                   | 160                   | 215                      |

Is there a difference in anaerobic thresholds among the different groups of athletes?

**Step 1.** Set up hypotheses and determine level of significance.

$H_0$ : There is no difference in median score of four population

Test with level of significance  $\alpha=0.05$

**Step 2.** Select the appropriate test statistic.

The test statistic for the Kruskal Wallis test is denoted  $H$  and is defined as:

$$H = \left( \frac{12}{N(N+1)} \sum_{j=1}^k \frac{R_j^2}{n_j} \right) - 3(N+1)$$

**Step 3.** Compute the test statistic.

| Distance Runners (A) |    | Distance Cyclists (B) |    | Distance Swimmers (C) |   | Cross-Country Skiers (D) |    |
|----------------------|----|-----------------------|----|-----------------------|---|--------------------------|----|
| 165                  | 3  | 178                   | 7  | 159                   | 1 | 180                      | 9  |
| 174                  | 6  | 181                   | 10 | 160                   | 2 | 187                      | 14 |
| 179                  | 8  | 182                   | 11 | 166                   | 4 | 195                      | 17 |
| 185                  | 13 | 190                   | 15 | 170                   | 5 | 201                      | 18 |

|     |                    |     |                    |     |                    |     |                    |
|-----|--------------------|-----|--------------------|-----|--------------------|-----|--------------------|
| 192 | <b>16</b>          | 209 | <b>19</b>          | 183 | <b>12</b>          | 215 | <b>20</b>          |
|     | R <sub>1</sub> =46 |     | R <sub>2</sub> =62 |     | R <sub>3</sub> =24 |     | R <sub>4</sub> =78 |

$$H = \left( \frac{12}{N(N+1)} \sum_{j=1}^k \frac{R_j^2}{n_j} \right) - 3(N+1) = \frac{12}{20(21)} \left( \frac{46^2}{5} + \frac{62^2}{5} + \frac{24^2}{5} + \frac{78^2}{5} \right) - 3(21) = 9.11$$

**Step 4. Compare with Critical Values of H**

**For the Kruskal Wallis**, if sample size is 5 or more in each group, the test statistic H approximates a chi-square distribution with df=k-1.

From Chi Square table critical value for df = 3 and α=0.5 is 7.81

Since  $9.11 \geq 7.81$ , Reject H<sub>0</sub>

**Intellectual Property** is a kind of property that is created by the human mind using his or her intellectual capability. Intellectual property is a form of intangible property, such as: 1) Copyright, 2) Patent, 3) Trademark, 4) Industrial Design, 5) Traditional Knowledge, 6) Trade Secret, 7) Geographical Indication, 8) Semiconductor Integrated Circuit Layout Design, 9) Developed Plant Varieties

Intellectual property rights (IPR) are rights of owner of IP.

Intellectual Property Rights are legal rights that are granted to the creator or owner of IP for a limited period of time and is subject to certain conditions.

IPR is an important topic for any research and innovation-oriented organization as well as academic and research institutes.

- It is important to protect IP.
- It is important to prevent infringement of third party's IP.
- Awareness of IPR gives direction to the research.
- IPR may be seen as revenue generation tool.
- Adds value and branding of an institute.
- Encourages and promotes entrepreneurship.

### **Types of Intellectual Property Rights (IPR)**

Intellectual Property Rights can be broadly divided into two categories:

- 1) Copyrights
- 2) Industrial Property

### **Copyright**

Copyright is a set of rights given by the law to the works of authorship. It is the right given to the creators of literary, dramatic, musical and artistic works and producers of cinematographic films.

Historically, copyright law British Statute of Anne 1709, was enacted for the printers of books. It was applied to the copying of books but later it applied to translations and derivative works. Presently, copyright covers a wide range of works, including maps, performances, paintings, photographs, sound recordings, motion pictures and computer programs.

The Indian Copyright Act, 1957 provides protection to following categories of works:

- 1) **Literary work:** books, articles, computer program, tables and compilations including databases
- 2) **Dramatic work:** drama, recitation, scenic arrangement & work capable of being performed by action
- 3) **Musical work:** Song, tune, beats
- 4) **Artistic work:** includes painting, sculpture, drawing (including diagram, map, chart), engraving, photograph, work of architecture; and any other work of artistic craftsmanship.
- 5) **Cinematographic work:** work of visual recording
- 6) **Sound recording:** means a recording of sounds from which sounds may be produced

## **Term of protection**

Literary, dramatic, musical and artistic works is 60-year from the year following the death of author.

For other works- 60-year from the date of publication.

Cinematograph films, sound recordings, photographs, posthumous publications, anonymous and pseudonymous publications, works of government and works of international organizations, the

## **Authorship**

Person who creates the work is the author of that work. Author has copyright.

If a work is "*made for hire*," employer, and not employee, is the author. An employer may be a firm, an organization, or an individual.

If an author has created the work while in his/her capacity of employee, the employer and not the employee is considered to be the author and copyright holder. The work done after office hours without directions of employer is not considered work for hire and author is owner of the work.

## **Patent**

*"The patent system added the fuel of interest to the fire of genius."* - Abraham Lincoln.

Patent originates from the Latin word *patere*, which means "*to lay open*". This indicates complete disclosure of the invention to the government is integral part of the process of patenting.

Patent is granted for invention to its creator.

## **Criteria of patentability**

The term invention is typically defined as 'A new product or process involving an inventive step and capable of industrial application'. An invention to be patentable, it must satisfy the criteria of patentability which is.

- a) Novelty
- b) *Inventive Step or Non-obviousness, and*
- c) *Industrial Application*

Grant of Patent gives certain rights to its holder by a state (national government) for a certain period, subject to certain conditions.

## **Aim of patent system**

The main aim of patent system is to:

- Encourage industrial development;
- Encourage complete disclosure of invention;
- Prevent duplication of research;
- Focus on commercial relevance of the invention;
- Prevent exploitation of researchers by giving them recognition as inventor and providing them royalty upon commercialization;
- Generate revenue when the invention gets commercialized; and
- Provide a source of technical information.

## **Criteria of patentability**

Well, the term invention is typically defined as '*A new product or process involving an inventive step and capable of industrial application*'. This means that for an invention to be patentable, it must satisfy the criteria of patentability which is: a) *Novelty*, b) *Inventive Step or Non-obviousness*, and c) *Industrial Application*

**Industrial application** does not mean it has to be related to manufacturing or machinery. It means, to be patentable, the invention must have a practical use and implementable in a real-world setting. This ensures that the invention is not just a theoretical idea but has potential commercial or practical value.

In India, the first patent statute was passed in 1857, but a proper Patents Act, was passed in 1970.

Indian Patent Office is administered under Ministry of Commerce and Industry with Head Offices at Kolkata and branch offices in Chennai, Mumbai and Delhi.

## **Trademark**

Trademark is one more form of IPR. It can be defined as a distinctive sign, which identifies certain goods or services as those produced or provided by a specific person or enterprise. Trademarks may be one or combination of words, letters, numerals, drawings, symbols, colours and shapes used packaging of goods as distinguishing feature.

A trademark:

- is used to identify and distinguish the goods/services
- is used to indicate the source
- is a symbol of good will
- guarantees consistency in quality
- makes purchasing decisions easy
- advertises & creates an image of a brand

Trademark is governed by Trademark Act. Trademark is said to be infringed if used in the course of trade or is printed and published in an advertisement.

The main purpose of a trademark is to enable consumers to identify / distinguish a product of a particular company from other similar products provided by competitors.

## **Industrial Designs**

An industrial design is an intellectual property right that protects the visual design of objects that are not purely utilitarian or useful.

Colourful handicrafts, exquisite piece of jewellery, unique looking handbag, design on clothings, etc.

## **Geographical indication**

Geographical indication or GI is an indication that is used to identify agricultural, natural or manufactured goods originating from a particular region. It has special quality or characteristics that are unique to that particular geographical region.

Some of the examples of Geographical Indications in India include Basmati Rice, Darjeeling Tea, Kancheepuram silk saree, Alphonso Mango, Nagpur Orange, Kolhapuri Chappal, Bikaneri Bhujia, etc.

## **Semiconductor IC**

Semiconductor IC is a product having transistors and other circuitry elements formed on a semiconductor material, which is designed to perform an electronic circuitry function. IPR is governed by The Semiconductor Integrated Circuits (IC) Layout Design Act 2000.

## **Trade Secret**

A trade secret refers to data or information relating to the business which is not generally known to the public and which the owner reasonably attempts to keep secret and confidential.

Trade secrets generally give the business a competitive edge over their rivals. Almost any type of data, processes or information can be referred to as trade secrets so long as it is intended to be and kept a secret, and involves an economic interest of the owner.

The terms, “confidential information” and “trade secret” are used alternatively and are usually confused with each other. Confidential information is the information which a company would like to keep secret; it may or may not be essential for the business. In other words, all trade secrets are confidential in nature but all confidential information cannot be termed as trade secret.

For any information to be considered as a trade secret:

- It must not be generally known or readily accessible by people who normally deal with such type of information.
- It must have commercial value as a secret
- The lawful owner must take reasonable steps to keep it a secret

There may or may not be any defined legal framework to protect ‘trade secret’ but misappropriation may be an offence.

## **Plant Variety and Farmer’s Rights**

The Protection of Plant Variety and Farmer’s Rights Act, 2001 provides for the establishment of an effective system for protection of plant varieties, the rights of farmers and plant breeders and to encourage the development of new varieties of plants. Using these rights, the farmers can grow new types of plants, crops which in turn improve the country’s economy.

Registration of a plant variety gives protection only in India and confers upon the rights holder, its successor, agent, or licensee the exclusive right to produce, sell, market, distribute, import, or export the variety.

## **Traditional Knowledge**

Traditional knowledge is knowledge, know-how, skills and practices that are developed, sustained and passed on from generation to generation within a community. Traditional knowledge also encompasses the wisdom, knowledge, and teachings of these communities. In many cases, traditional knowledge has been orally passed for generations from person to person. Some forms of traditional knowledge are expressed through stories, legends, folklore, rituals, songs, and even laws.

Traditional knowledge can be found in a wide variety of contexts, including: agriculture, technology, medicine, wellness, etc. However, traditional knowledge or knowledge that has ancient roots is often oral and it is not protected by conventional intellectual property systems.



## Copyright

A copyright is a form of intellectual property, which protects original works of authorship such as literary (books, periodicals, computer Programs, tables and compilations), dramatic, musical, artistic, cinematographic and sound recording, etc.

It is a bundle of rights; (i) rights of reproduction, (ii) communication to the public, (iii) adaptation, and (iv) translation of the work.

## International Treaty on Copyright

The **Berne Convention for the Protection of Literary and Artistic Works** is an international agreement that was adopted in 1886 in Bern, Switzerland. It has been revised from time to time. Its aims to protect the rights of authors of literary and artistic works.

### Scope

- **Works Covered:** The convention covers literary and artistic works, including books, music, paintings, sculptures, and films.
- **International Recognition:** Member countries must recognize the rights of authors from other member countries as if they were their own nationals.

### Membership

- **Global Reach:** As of now, the Berne Convention has been ratified by 181 countries.
- **World Trade Organization (WTO):** Only countries that are signatories to the Berne Convention can qualify to be members of the WTO.

India is a member of this treaty.

### Key Principles

1. **Automatic Protection:** Protection is granted automatically without the need for registration or formalities.
2. **Minimum Standards:** The convention sets minimum standards of protection that member countries must provide to works of authors from other member countries.
3. **Moral Rights:** Authors have the right to claim authorship of their work and to object to any distortion, mutilation, or other modification that would harm their honor or reputation.

Points under this Treaty:

Registration of copyright is not an essential requirement.

The moment an original work is created, it is automatically copyrighted to the author and is simultaneously protected in all the member countries of the Berne Convention.

Copyright Act, 1957 governs the copyright protection in India.

## Requirements For Copyright Protection

The copyright law protects the expression of the idea and not the idea. Apart from this, the work shall be original and fixed in a tangible form.

### **i. Originality**

In order to get protection as copyright, the work has to be original and should not be copied from any source.

### **ii. Fixation**

The work shall be fixed in a tangible form or reduced to material form, recorded, written, painted or typed. A work is considered fixed when it is stored on some medium in which, it can be perceived, reproduced, or otherwise communicated. For example, a story is considered fixed when it is written down on paper. The paper is the medium on which the story can be perceived, reproduced and communicated. Similarly, a song is fixed when it is recorded, a computer program code is fixed when it is typed or written.

### **iii. Expression of idea and not idea itself**

Copyright law does not protect the ideas but only expression of the idea. It is sometimes very challenging to draw a line between idea and expression.

**Term of Protection:**

60 years from death of author: literary, dramatic, musical and artistic works

60 years from date of publication: films, sound recording, pseudonymous publications,  
work by any kind of organization

**Registration of Copyright:**

Register of Copyrights is maintained in the Copyright Office of the Department of Education.

1. Prepare the Work: Ensure that the work is original and fixed in a tangible form (e.g., written, recorded, or saved digitally).
2. Complete the Application: Fill out Form XIV, which is the application form for copyright registration. One can submit this form online through the official Copyright Office website or offline by visiting the office.
3. Pay the Filing Fee: Submit the required filing fee along with application. Fees can vary depending on the type of work and the method of submission.
4. Submit a Copy of Work: Mail or upload a physical or digital copy of the work to the Copyright Office. This completes the submission process for registration.
5. Wait for Confirmation: The Copyright Office will review the submission and, if everything is in order, issue a certificate of registration.

### **Why Registration is Recommended**

1. The certification of copyright registration is an authentic proof of valid title and ownership.
2. Copyright registration establishes a public record of the copyright claim.
3. It is helpful for assignment or licencing of copyright.

**Notice of Copyright:**

Notice of copyright is an identifier placed on copies of the work. It is optional. It does not require advance permission from, or registration with the Copyright Office. Copyright notice informs the public that the work is protected by copyright, identifies the copyright owner, and shows the year of first publication of the work.

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### Who Owns the Copyright in a Work?

- In the case of a literary work the author
- In the case of a musical work, the composer.
- In the case of a cinematograph film, the producer.
- In the case of a sound recording, the producer.
- In the case of a photograph, the photographer.
- In the case of a computer-generated work, the person who causes the work to be created

### Penalties for Copyright Infringement

In India, for example, the **Copyright Act, 1957** outlines the penalties for copyright infringement.

If someone is found guilty of copyright infringement, they could face:

- **Imprisonment:** A minimum of six months, which can extend up to three years.
- **Fines:** A minimum fine of Rs. 50,000, which can go up to Rs. 2 lakhs.

These penalties are meant to deter unauthorized use of copyrighted works and protect the rights of the copyright holders.

### What Is Copyright Infringement?

Anyone who exploits any of the exclusive rights of copyright without the copyright owner's permission commits copyright infringement. Infringement could be either literal or non-literal. Literal infringement is the exact copying of original work and it can be verified by comparing original and copied work, while "non-literal" copying is paraphrased, or loosely paraphrased.

In order for a court to determine that a copyright in a work has been infringed, it must find that:

- (1) The infringing work is "substantially similar" to the copyrighted work
- (2) The alleged infringer had access to the copyrighted work
- (3) There are no clear rules for deciding what is "substantial similarity". Courts look for similarities in appearance, sound, words, format, layout, sequence, and other elements of the works.

### What is Fair Use in Copyright

Fair use is a legal doctrine that allows limited use of copyrighted material without needing to obtain permission from the copyright owner. It's designed to balance the interests of copyright holders with the public's need to access and use creative works.

Fair use include:

1. **Research and Education:** Using copyrighted material for educational purposes, such as in a classroom or for academic research.
2. **Criticism and Review:** Quoting parts of a work for criticism, review, or commentary, provided it is properly attributed.
3. **News Reporting:** Using portions of a work in news reporting to inform public.
4. **Parody and Satire:** Creating a parody or satire that transforms the original work in a way that is considered fair use.
5. **Personal Use:** Making a copy of a work for personal, non-commercial use.

### Copyright Violation and Plagiarism

**Copyright Violation** involves the unauthorized use and distribution of copyrighted material.

**Plagiarism** is the act of presenting someone else's work, ideas, or expressions as your own without proper acknowledgment.

In India, plagiarism can lead to both civil and criminal penalties under the Indian Copyright Act, 1957. Here are the key legal penalties:

#### Civil Penalties

- **Damages and Injunctions:** Under Section 57, the author can claim damages and seek an injunction if their work is distorted, mutilated, or modified in a way that harms their honor or reputation.

#### Criminal Penalties

- **Imprisonment:** Under Section 63, plagiarism can lead to imprisonment for a term of six months to three years.
- **Fines:** The offender may also be fined between ₹50,000 and ₹2,00,000 (approximately \$625 to \$2,500).
- **Repeat Offenses:** For repeat offenses, the penalties are more severe, with imprisonment ranging from one to three years and fines between ₹1,00,000 and ₹2,00,000 (approximately \$1,250 to \$2,500).

Plagiarism is taken seriously in both academic and professional settings, and it's important to always give proper credit to the original authors to avoid these penalties.

Owner of copyrighted work can transfer his rights to anyone by means of:

- Assignment
- License

Assignment is a permanent transfer of rights where license agreement can be executed for a specified duration of time. The assignment shall be in writing signed by the assignor (copyright owner) or by his duly authorised agent. It shall mention the specific works and specify the rights assigned and the duration and territorial extent of such assignment. It shall also specify the amount of royalty payable to the author or his legal

heirs during the currency of the assignment and the assignment shall be subject to revision, extension or termination on terms mutually agreed upon by the parties. If the period of assignment is not stated, it shall be deemed to be five years from the date of assignment. If the territorial extent of the assignment of the rights is not specified, it shall be presumed to extend within the whole of India.

### **Difference between Plagiarism, Copyright Infringement, Piracy**

#### **Plagiarism**

- **Definition:** Presenting someone else's ideas, words, or work as your own without proper attribution.
- **Nature:** Ethical/academic misconduct.
- **Scope:** Covers ideas, research, and text even if not protected by copyright.
- **Example:** Copying a paragraph from a thesis or article, rephrasing it, and submitting it as your own without citation.
- **Consequence:** Academic penalties (expulsion, degree cancellation, blacklisting). Not always a legal offense unless copyright is also violated.

#### **Copyright Violation (Copyright Infringement)**

- **Definition:** Unauthorized use of someone's copyrighted work (literary, artistic, musical, software, etc.).
- **Nature:** Legal offense under copyright law.
- **Scope:** Protects the *expression* of ideas, not the ideas themselves.
- **Example:** Copying and publishing a book chapter without permission.
- **Consequence:** Civil lawsuits, damages, fines, and imprisonment (in India, up to 3 years under the Copyright Act, 1957).

#### **Piracy**

- **Definition:** Large-scale, often commercial, unauthorized reproduction and distribution of copyrighted works.
- **Nature:** Criminal offense.
- **Scope:** Typically involves movies, music, software, or books distributed illegally.
- **Example:** Selling pirated DVDs, uploading cracked software, or distributing movies on torrent sites.
- **Consequence:** Severe penalties — fines, imprisonment, and seizure of goods.

### **Quick Comparison Table**

| <b>Aspect</b>     | <b>Plagiarism</b>                 | <b>Copyright Violation</b>                        | <b>Piracy</b>                                 |
|-------------------|-----------------------------------|---------------------------------------------------|-----------------------------------------------|
| <b>Definition</b> | Using others' work without credit | Unauthorized use of copyrighted work              | Illegal mass distribution of copyrighted work |
| <b>Nature</b>     | Ethical/academic misconduct       | Legal offense                                     | Criminal offense                              |
| <b>Scope</b>      | Ideas, text, research             | Expression of ideas (books, art, music, software) | Large-scale reproduction/distribution         |
| <b>Example</b>    | Copying thesis                    | Publishing book                                   | Selling pirated                               |

| Aspect      | Plagiarism         | Copyright Violation        | Piracy               |
|-------------|--------------------|----------------------------|----------------------|
|             | without citation   | chapter without permission | movies/software      |
| Consequence | Academic penalties | Civil/criminal penalties   | Criminal prosecution |

In short:

- **Plagiarism** = dishonesty in attribution.
- **Copyright violation** = breaking the law by using protected work without permission.
- **Piracy** = large-scale illegal copying/distribution of copyrighted material.

AI Generated Text

### Copyright Infringement

- **AI-generated text** is generally considered **machine-created content**.
- Copyright law protects works created by **human authors**. In most jurisdictions (including India, US, EU), purely AI-generated works **do not qualify for copyright protection** because there is no human author.
- Publishing AI-generated writing **does not infringe copyright**, unless:
  - The AI output **copies substantial portions of copyrighted material** (e.g., reproducing a book passage verbatim).
  - The AI was trained on copyrighted works and outputs something nearly identical.

So, if the AI produces original text, publishing it is **not copyright infringement**.

### Plagiarism

- Plagiarism is about **academic honesty and attribution**, not law.
- If someone publishes AI-generated writing and **pretends it is their own original thought/research**, it can be considered **plagiarism** in academic or professional contexts.
- Universities and journals often require disclosure if AI tools were used. Failure to disclose may be treated as misconduct.

So, publishing AI-generated text **without attribution or disclosure** can amount to **plagiarism**, even if it's not copyright infringement.

### Key Takeaway

- **AI-generated + original** → Not copyright infringement.
- **AI-generated + no disclosure** → Can be plagiarism in academia/professional writing.
- **AI-generated + copies existing work** → Both plagiarism *and* copyright infringement.

### When AI violates Copyright

There are already real-world lawsuits where AI outputs have been accused of **copyright infringement**. Courts are actively examining whether AI companies or users are liable when generative models reproduce existing works.

### Key Examples of AI Copyright Cases

- **Andersen v. Stability AI (2023)**
  - Filed by a group of visual artists against *Stable Diffusion*.
  - Claim: The AI model allegedly copied and reproduced artists' copyrighted images without permission.
  - Status: Still ongoing, but it set the stage for many similar lawsuits [Copyright Alliance](#).
- **Getty Images v. Stability AI (UK, 2023–2025)**
  - Getty accused Stability AI of scraping millions of copyrighted photos to train its model.
  - Claim: Unauthorized use of Getty's library amounted to copyright infringement.
  - Status: UK trial reached closing stages in 2025, with courts scrutinizing whether training data use counts as infringement [Publishers Association](#).
- **Media Companies vs. OpenAI, Microsoft, Google, Anthropic (2024–2025)**
  - Multiple publishers and news outlets filed suits claiming AI models were trained on copyrighted articles and sometimes reproduced them.
  - Example: The *New York Times* sued OpenAI and Microsoft in late 2023, alleging ChatGPT outputs contained near-verbatim passages from its reporting [sustainabletechpartner.com](#).
- **Anthropic Case (US, 2025)**
  - Court found that downloading books from pirate sites to train AI was **not fair use**, but scanning library books for training was allowed.
  - Shows courts are distinguishing between lawful and unlawful training sources [Publishers Association](#).

### What These Cases Show

- **AI itself cannot be sued** (not a legal entity). Responsibility falls on **users** (who publish infringing outputs) or **developers** (who train on copyrighted material).
- **Training vs. Output:** Courts are debating whether using copyrighted works in training is infringement, but reproducing them in outputs almost certainly is.
- **Global Impact:** Lawsuits in the US, UK, and EU are shaping how copyright applies to AI, with no final global consensus yet.

### Takeaway

- If AI generates **original content**, publishing it is safe.
- If AI generates **existing copyrighted work**, publishing it can expose the **user** (and sometimes the developer) to copyright liability.

- Courts are still deciding how far copyright law extends to AI training and outputs, but the trend is clear: **verbatim reproduction = infringement**.



## **What is a Patent?**

A Patent is a statutory right for an invention granted for a limited period of time to the patentee by the Government, in exchange of full disclosure of his invention for excluding others.

Term of a patent: 20 years

The patent system in India is governed by the Patents Act, 1970 (No.39 of 1970) as amended from time to time.

Patent is country specific. Patents should be obtained in each country where the applicant requires protection of his invention.

Separate patent is needed for each country.

## **What can be patented?**

An invention relating either to a product or process that is new, involving an inventive step (non-obviousness) and capable of industrial application can be patented. However, certain categories of inventions are non- patentable.

### **Inventions that are not patentable in India**

- 1) an invention which is frivolous or which claims contrary to natural laws;
- 2) an invention contrary to public order or morality or harmful
- 3) discovery of scientific principle, living thing or non-living substance occurring in nature
- 4) discovery of a new form of a known substance
- 5) a substance obtained by mere admixture resulting only in the aggregation of the properties of the components
- 6) mere arrangement or re-arrangement or duplication of known devices each functioning independently of one another in a known way;
- 7) a method of agriculture or horticulture;
- 8) any process for medicinal, surgical, curative, prophylactic (diagnostic, therapeutic) or other treatment of human beings or animals
- 9) plants (including seeds) and animals other than microorganisms;
- 10) a mathematical or business method or a computer program;
- 11) a literary, dramatic, musical or artistic work or any other aesthetic creation whatsoever including cinematographic works and television productions;
- 12) etc...

## **Application for a Patent**

Application can be filed at Indian Patent Office or online.

The information relating to the patent application is available in website of the Patent Office, [www.ipindia.gov.in](http://www.ipindia.gov.in)

A patent application can be filed either by first inventor or his assignee, either alone or jointly with any other person. Legal representative of any deceased person can also make an application for patent.

A patent application can be filed with provisional specification or with complete specification along with fee as prescribed in schedule.

An application for patent can be filed either in Hindi or English.

The types of applications

#### **A) Ordinary Application**

An application for patent filed in the Patent Office without claiming any priority either in a

convention country or without any reference to any other earlier application under process in the office. Such type of application is known as an ordinary application.

Ordinary application can be filed by way of filing

(i) Provisional specification, or (ii) Complete specification:

#### **B) Convention Application**

An application for patent filed in the Patent Office, claiming a priority date based on the same or substantially similar application filed in one or more of the convention countries is known as a convention application.

#### **C) PCT International Application**

An Application filed in India under Patent Cooperation Treaty is an international application which can be considered as deemed filed in more than 150 countries (member of PCT) by a single application.

#### **D) PCT National Phase Application**

When an international application is made according to PCT designating India, an applicant can file the national phase application in India within 31 months from the international filing date or the priority date, whichever is earlier.

#### **E) Patent Of Addition**

where an application is made for a patent in respect of any improvement in or modification of an invention.

#### **F) Divisional Application**

Dividing an invention into parts,

### **Patent Publication**

Every application for patent is published before it is granted.

Application forms: filing up a set of forms is required.

## List of Forms to be Submitted

- Form 1 : Application for grant of patent
- Form 2 : Provisional / complete specification
- Form 3 : Statement & undertaking under section 8
- Form 5 : Declaration as to inventorship
- Form 9 : Request for publication
- Form 18 : Request for examination of application for patent

A complete specification contains the following:

Title, Technical field of the invention, Background of the invention, Objects of the invention, Statement of the invention, A brief description of the drawings, A detailed description of the invention, Claims, Abstract

## Patent Search

- Indian Patent <https://iprsearch.ipindia.gov.in/publicsearch>

## Fees

| FORMS   | APPLICANT-NATURAL | APPLICANT-OTHER THAN NATURAL PERSON |                            |
|---------|-------------------|-------------------------------------|----------------------------|
|         |                   | SMALL ENTITY                        | OTHERS EXCEPT SMALL ENTITY |
| FORM 1  | 1600              | 4000                                | 8000                       |
| FORM 2  | NO FEES           | NO FEES                             | NO FEES                    |
| FORM 3  | NO FEES           | NO FEES                             | NO FEES                    |
| FORM 5  | NO FEES           | NO FEES                             | NO FEES                    |
| FORM 9  | 2500              | 6250                                | 12500                      |
| FORM 18 | 4000              | 10000                               | 20000                      |

## LIFE CYCLE OF A PATENT APPLICATION

### Patent Opposition

In India, both **pre-grant opposition** and **post-grant opposition** are mechanisms that allow third parties to challenge the grant of a patent.

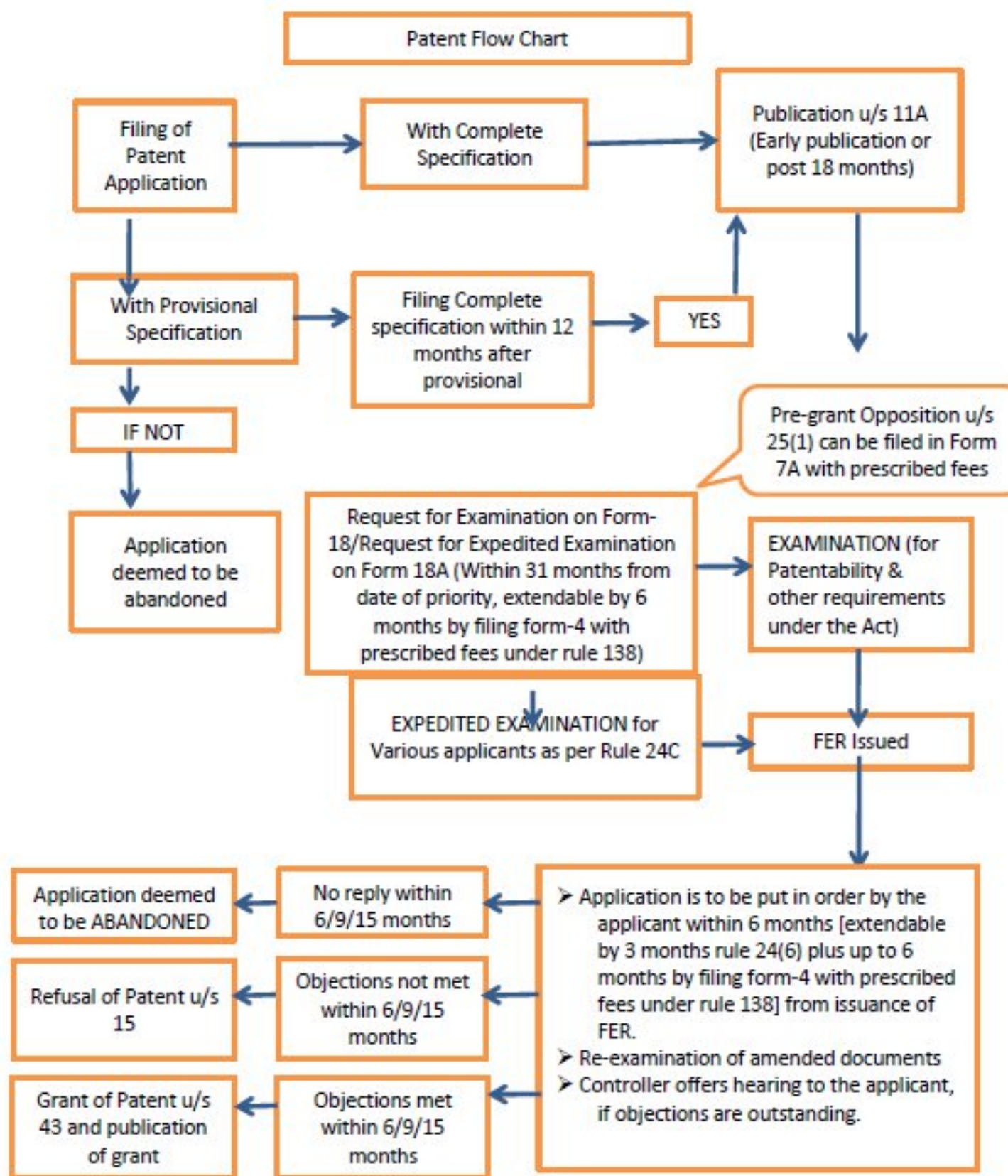
### Pre-Grant Opposition

**Timing:** Filed after publication of the patent application but before grant of patent.

**Who Can File:** Any person can file a pre-grant opposition.

**Grounds:** The grounds for opposition include the invention being publicly known, obvious, not patentable, or the applicant not disclosing relevant information.

**Procedure:** The opposition is filed in writing. The Controller examines the statements of opponent and applicant before granting the patent.



## Post-Grant Opposition

Patent can be challenged even after it is granted but within one year from the date of publication of the grant.

### **Patent Agent:**

A Patent agent is a person registered with Indian Patent Office who is entitled—

- (a) to practice before the Controller General of Patent, Trademark and Design; and
- (b) to prepare all documents, transact all business and functions in connection with any proceeding before the Controller under this Act.

Eligibility – citizen of India; completed the age of 21 years; has a degree in science, engineering or technology; must pass an examination conducted by patent office.

It is not necessary to engage a registered patent agent for filing an application for patent.

### **Types of Patent**

- Utility Patents
- Design Patents
- Plant Patents

### **Key Differences between Utility and Design Patent**

**Legal Acts:** Utility Patents are governed by Patents Act, 1970 whereas Design Patents are administered by the Designs Act, 2000.

**Nature of Protection:** Design patents protect the visual appearance of an article, while utility patents protect the functional aspects of an invention.

**Examination Criteria:** Design patents are examined for novelty and originality, whereas utility patents are examined for novelty, inventive step, and industrial applicability.

**Duration:** Design patents term is 10 years, while utility patents last for 20 years.

There is no provision of Plant Patents in India. This aspect is governed by The Protection of Plant Varieties and Farmers' Rights (PPVFR) Act, 2001.

### **Obligations of Patentee**

- Encourage and secure working of invention to fullest extent without undue delay.
- Not to use monopoly to unfairly prejudice interest of public.
- Failure to discharge duties reasonably, would lead to revocation of patent.
- To furnish periodical statements to Controller in Form 27, as to extend to which invention has been worked. Failure to provide statement to Controller may lead to imposition of penalty.
- Patentee has to maintain the patent by paying renewal fee after two years onwards. Patentee renewal fee can be paid every year or lumpsum.

### **Patent Infringement**

There are civil and criminal consequences.

## Civil Consequences

1. **Injunctions:** The patentee can seek an interim injunction to stop the infringing activities immediately. After a full trial, a permanent injunction may be granted.
2. **Damages:** The patentee can claim monetary compensation for the losses.
3. **Accounting of Profits:** The infringer may be required to account for and surrender the profits made from the infringement.

## Criminal Consequences

1. **Fines and Imprisonment:** Claiming patent rights without authorization is a criminal offense. Penalties can include fines and imprisonment.

## Remedies

1. **Administrative Remedies:** The patentee can approach the collector of customs to prevent the importation of infringing goods into India.
2. **Legal Action:** The patentee can file a lawsuit in the appropriate jurisdiction where the infringement occurred.

## Timeline for obtaining a patent in India

1. **Filing the Application:** The process begins with filing a provisional or complete patent application. A provisional application can be filed without a set of claims, while a complete application includes detailed claims<sup>1</sup>.
2. **Early Publication (Optional):** You can request early publication of your application by paying the prescribed fee. If not requested, the application will be published 18 months from the priority date<sup>1</sup>.
3. **Request for Examination:** This must be filed within 48 months from the priority date. The examination process will not begin until this request is made<sup>1</sup>.
4. **First Examination Report (FER):** Once the examination begins, the patent office issues an FER if there are objections. You have 12 months to respond to the FER<sup>1</sup>.
5. **Hearing and Final Decision:** After addressing the objections, the patent office may schedule a hearing. The final decision (grant or rejection) is made based on the responses and any further clarifications provided.
6. **Grant of Patent:** If all requirements are met, the patent is granted, and a certificate of patent is issued.

## Approximate Timeline:

- **Filing to Publication:** 18 months (or earlier if early publication requested)
- **Publication to Grant:** 12-24 months (depending on the complexity and any objections raised)

## Research Reports

The research report is a concrete proof of the study that was undertaken.

The significant role that a research report is as follows:

- 🎬 documents all steps from framing of research question to interpretation of results and study findings
- 🎬 details of how and why the steps were carried out, i.e. justification for choosing one technique over the other.
- 🎬 authenticates the quality of the work and establishes the strength of the findings
- 🎬 gives direction in terms of implication of results
- 🎬 serves as a framework for further research on same area

## Types of Research Reports

| Types of Research Reports                                                                 |                                                                                                |
|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Brief Reports                                                                             | Detailed reports                                                                               |
| <ul style="list-style-type: none"><li>• Working papers</li><li>• Survey reports</li></ul> | <ul style="list-style-type: none"><li>• Technical reports</li><li>• Business reports</li></ul> |

### Brief Reports

These kinds of reports are not formally structured and are generally short, usually about 4 -5 pages.

**Working papers** or basic reports are written for the purpose of recording the research process in terms of scope of study, framework of study, methodology used, results and findings. However, interpretation of findings, study background, and past literature is usually not included.

**Survey reports:** Might not have an academic orientation. The focus is to present in simple and easy to understand format that includes figures and tables.

### Detailed reports

These are more formal and could be academic, technical or business reports.

🎬 **Technical reports:** These are major documents and would include all elements of the basic report, as well as the interpretations and conclusions. All sources of data, sampling plan, data collection instrument(s), data analysis outputs are formally and sequentially documented.

🎬 **Business reports:** These reports include conclusions as understood by the business manager. Focus is more on narrative form to facilitate decision making by using pictorially representation, bar charts and graphs. Tabular data might be attached in the appendix.

## Structure of Research Report

Research report has a structured format, and follows a standardized process. It is divided into standard sections and sub-sections.

Detailed technical report contains all elements of research report.

|                                   |  |
|-----------------------------------|--|
| Preliminary Section               |  |
| Title Page                        |  |
| Letter of Authorization           |  |
| Executive Summary                 |  |
| Acknowledgements                  |  |
| Table of Contents                 |  |
| Background Section                |  |
| Problem Statement                 |  |
| Study Introduction and Background |  |
| Scope Objectives of the Study     |  |
| Review of Literature              |  |
| Methodology Section               |  |
| Research Design                   |  |
| Sampling Design                   |  |
| Data Collection                   |  |
| Data Analysis                     |  |
| Findings Section                  |  |
| Results                           |  |
| Interpretation of Results         |  |
| Conclusions Section               |  |
| Conclusion Recommendations        |  |
| Limitations of the Study          |  |
| Appendices                        |  |
| Glossary of Terms                 |  |
| Bibliography                      |  |

The sequencing may be reversed in the management report to suit the needs of the decision-maker.

### **Preliminary Section**

This section mainly consists of identification information of the study. It has the following individual elements:

**Title page:** The title should be crisp and indicative of the nature of the project.

**Letter of Transmittal:** This is the letter that broadly refers to the purpose of study.

**Letter of Authorization:** Copy of the letter that authorised the study.

**Executive summary:** The summary of the entire report is presented in narrative form.

**Acknowledgements:** A small note acknowledging the contribution of all those who provided inputs for accomplishing the study.

**Table of contents:** List major sections and subsections of the study, list of tables, list of figures, graphs, exhibits (if any) and list of appendices with page numbers.



## **Main Report**

This is the most significant and academically robust part of the report. It is divided into standard sections.

**Background Section:** It has following sub-sections, namely, Problem Statement, Introduction and Study Background, Scope and Objectives of Study, and Review of Literature

- **Problem Definition/ Statement:** This sub-section presents the formal definition of the research problem.
- **Study background:** Study background gives a description of the theoretical and contemporary data that led to research problem and guided the research.
- **Study scope and objectives:** it presents statements related to the purpose of the study and the formulated hypotheses.
- In case of academic research, there is a separate section devoted to the review of related literature, which presents a detailed reporting of work done on the same or related topic of interest.

**Methodology of Research:** The section would essentially have five to six sub-sections specifying the details of how the research was conducted. These would essentially be:

- **Research framework or design:** The variables and concepts being investigated are clearly defined, with a clear reference to the relationship being studied. The justification for using a particular design also has to be presented here.
- **Sampling design:** The entire sampling plan in terms of the population being studied, along with the reasons for collecting the study-related information from the given group is given here.
- **Data collection methods:** Lists the information needed for the study as drawn from the study objectives stated earlier. The secondary data sources considered and the primary instrument designed for the specific study are discussed here.
- **Data analysis:** The assumptions and constraints of the analysis are explained here in simple, non-technical terms.

**Study results and findings:** This is the most critical chapter of the report.

**Results:** Presents the collected data and its analysis. This section often includes tables, graphs, and descriptive statistics to illustrate the findings.

**Interpretation of Results:** linking analysis of data to the research questions and existing literature.

**Conclusion:** Summarizes the key findings and their implications, Limitations of the study and may suggest areas for future research.

**Appendices:** May consist of two kinds of information.

1. **Secondary information:** Technical information in form of article that needs to be understood by the reader; long tables, legal or policy documents.

2. Primary data that cannot be compressed and presented in the main body of the report. This includes original questionnaire, formula used for the study, Computer programs used in study, long tables and graphs which can be described in statement form in the text.

**Bibliography:** Lists the papers cited in a standardized format. Lists may be presented in order of occurrence in text or in alphabetical order based on last name of author.

**Glossary of Terms:** In case there are specific terms and technical jargon used in the report, it is put in a glossary. This section is usually the last section of the report.

## Research Proposal

Research proposal is similar to research report. Typical outline of Research proposal is as under.

Here's a general outline you might consider following:

1. **Title Page:** Includes research title, name of researcher(s), and their institutional affiliation.
2. **Abstract:** Contains a brief summary of research proposal, including the research question, objectives, methodology, and potential significance.
3. **Introduction:**
  - **Background:** Contextualize the research problem.
  - **Problem Statement:** States the problem that research will address.
  - **Research Questions:** Lists the primary and any secondary questions that the proposed research aims to answer.
  - **Objectives:** Defines the aims of proposed research.
4. **Literature Review:** Contains the summary of existing research related to research topic. Identify gaps that the research will fill.
5. **Research Methodology:**
  - **Research Design:** Explains the overall approach (e.g., qualitative, quantitative, mixed methods) that will be used.
  - **Data Collection:** Describes how and what data will be collected.
  - **Data Analysis:** Explain how the data will be analysed.
  - **Ethical Considerations:** Address ethical issues related to proposed research if any.
6. **Timeline:** Provide a detailed schedule of research activities. This is an important section of research proposal. It stipulates how the progress in research is evaluated.
7. **Budget:** Estimate the costs associated with research and justify the budget.
8. **References:** List all the sources cited in research proposal.
9. **Appendices** (if applicable): Include any additional materials, such as survey instruments, interview guides, etc.